K–8 Rightsizing & Consolidation Scenario Modeling

Technical Documentation

Portland Public Schools Data and Accountability · October 2026

About this document: This document gathers in one place the technical description of how the rightsizing scenarios are modeled. It combines four documents written during the work. Part 2, the Enrollment Methodology, is kept as the Portland State University Population Research Center (PRC) reviewed it, apart from its title and references to other parts. The other three were edited for readability. An overview and two short parts, on how scenarios are evaluated and on the underlying assumptions, were added.

A note on names and figures: The examples in Parts 2, 3 and 4 use generic names (School A, Boundary Area 1) and simplified figures. Real schools are named only where a part describes a rule or a published result.

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Part 1. Overview

The rightsizing scenarios are evaluated with two models, both of which are built upon the same data. The enrollment model (Part 2) starts from PRC's forecast and applies each scenario's rules to produce enrollment by school, grade and year through the 2035-2036 school year. The demographic impact model (Part 5) places each currently enrolled student under each scenario to show who would be affected. Both the enrollment model and demographic impact model read the same boundary work (Part 4).

1. What goes into a scenario

Most of what a scenario rests on are measured inputs, with two involving planning assumptions (see Part 7). Main inputs:

Input	What it is	Measured or assumed	Described in
PRC enrollment forecast	Students by school, residency group, grade and school year, 2026-2027 through 2035-2036	PRC's forecast, from its own methods	Part 2: Section 2.1
Boundary shares	The share of a school's residents who live in the census blocks a proposal reassigns	Measured. student counts by block, checked against eleven school years	Part 4: Sections 3 to 5
Residency table	Where each school's and program's students live, by current attendance area	Measured: three school years of geocoded enrollment	Part 2: Section 2.3, Part 4: Section 5.2
Follow shares	The share of transfer and immersion students who follow a closed school or a moved program	Planning judgment	Part 7: Section 1
Capture rate	The share of an area's resident families who attend its neighborhood school	PRC's rate for the current school, held unchanged at the receiving school	Part 7: Section 1
Enrollment Floor and Enrollment capacity	The minimum enrollment for each type of school, and each building's functional capacity	District thresholds; the district's school utilization model	Part 6

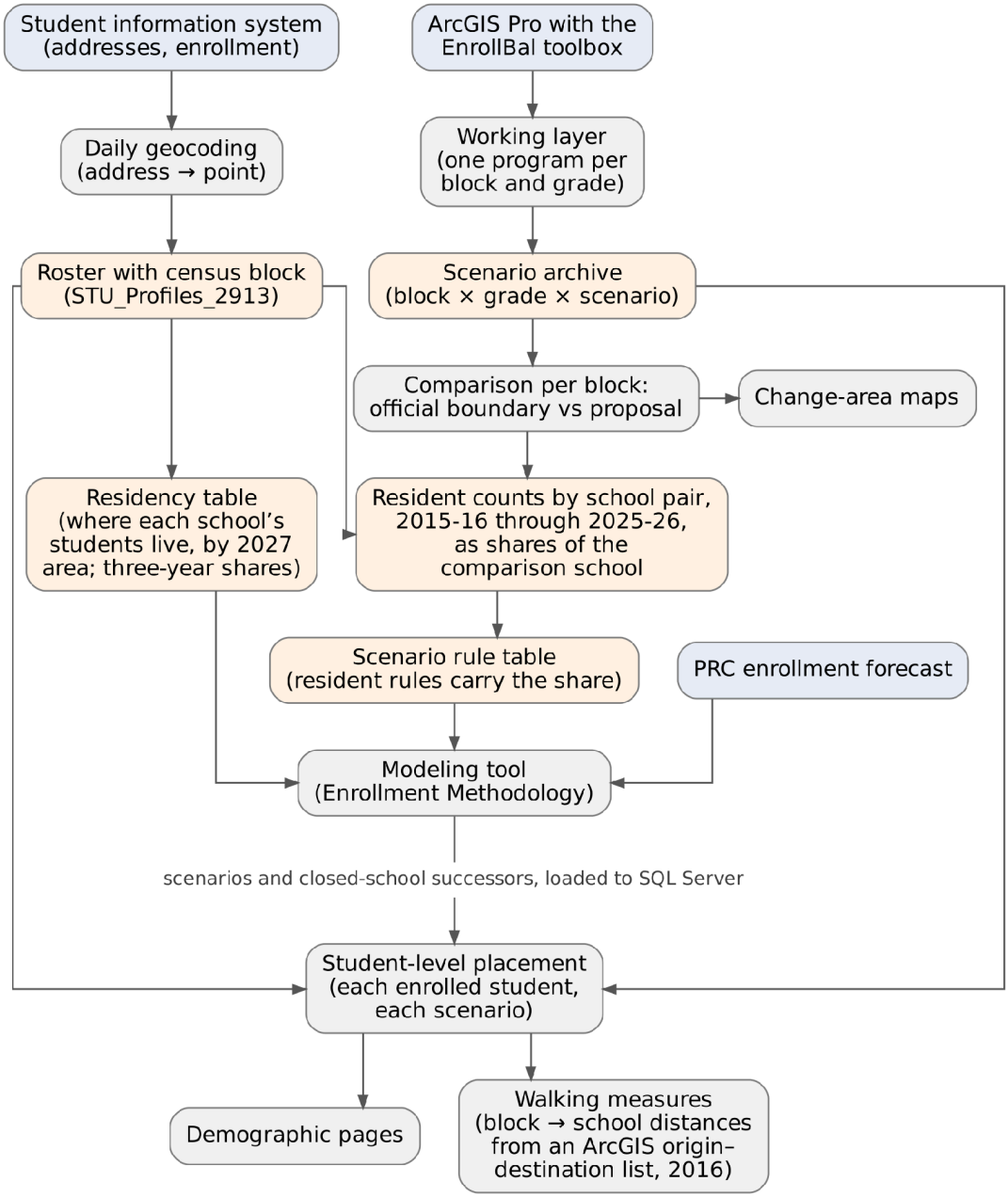


Figure 1. How the pieces connect. The modeling tool at the center of the diagram is the core scenario development tool. The diagram shows how information flows into the modeling tool and how it connects to the other component processes and scenario outputs. Starting at top left: addresses become a roster with a census block for every student; the roster also yields the residency table. Starting at the center top: a proposal drawn in ArcGIS Pro becomes block-by-grade assignments in the scenario archive. The two meet in the counting views, whose shares go into the scenario rule table. The rule table, the residency table and PRC's forecast feed the modeling tool. The tool's scenarios and closed-school successors are loaded into SQL Server, where, with the archive and the roster, they drive the student-level placements behind the demographic pages and walking measures.

2. What the scenario numbers are for

The scenario numbers compare proposals on a common basis: which schools reach their enrollment floors in 2031-2032, which exceed their capacity in any year, how many students move and to where, and how student groups and walking distances change.

Scenario numbers are not the enrollment forecast that staffing rests on. After the Board adopts boundaries, grade configurations and program locations, PPS will give PRC the October 2026 enrollment together with the adopted changes. PRC, using its own methods, will produce its own forecast, and the PRC result becomes the one-year forecast for staffing in 2027-2028.

3. Names and conventions

- **Scenarios.** The two scenarios compared throughout are Scenario A and Scenario B. Status Quo is PRC's forecast with no scenario applied.
- **Years.** A year written alone is the Fall of the school year. 2027 is the 2027-2028 school year. The first year of change in Scenarios A and B is 2027-2028.
- **Student groups.** Students are grouped as PRC forecasts them: residents attending the neighborhood program, non-residents attending it (transfer and lottery students), each language immersion program, and special programs. Part 2, Section 2.1 defines them.
- **The 2027 boundaries.** Every scenario is drawn against the 2027 boundary set: today's attendance areas with the one change already adopted for 2027-2028, the end of a dual assignment at one high school.

Part 2. Enrollment Methodology

How Portland Public Schools Combines PRC's Enrollment Forecasts with Proposed Planning Scenarios

Prepared by Portland Public Schools Data and Accountability · Reviewed by the Portland State University Population Research Center (PRC) · October 2026

A note on the examples in this part: PPS is currently evaluating preliminary consolidation scenarios in which the specific schools involved may still change. To keep this methodology usable on its own — and appropriate for wider sharing — every example below uses generic placeholder names ("School A," "Boundary Area 1," and so on) and simplified figures rather than real projected counts. The method illustrated is identical to what runs on real data; only the labels and numbers have been replaced.

What changed in this revision: This revision supersedes the August 2026 draft and incorporates PRC's review. Five things changed materially, each described in the section noted: how the tool interprets a percentage below 100 versus a percentage of exactly 100 (Section 3.5); the adjustment of the residency table when a scenario closes a school outright (Section 3.3); the consolidation of all rounding for one school, program, grade and year into a single step (Section 3.7); the replacement of the earlier constrained-optimization solver with an exact, reproducible equivalent (Section 3.7); and the addition of an automatic verification report (Section 5). Section 6 is new and explains how to read the tool's detailed output.

1. Purpose and Scope

Each year, the Population Research Center (PRC) at Portland State University provides Portland Public Schools (PPS) with independent, ten-year enrollment forecasts for every school, program, and grade in the district. PPS's planning process then asks a further question: what would enrollment look like under a proposed change — a school consolidation, a boundary adjustment, or a program merger?

To answer that question consistently and transparently, PPS built a modeling tool that takes a proposed change (a "scenario") and mathematically combines it with PRC's forecast to produce a revised, school-by-school and grade-by-grade projection. This part explains, in plain terms, exactly how that combination is done — the rules the tool follows, and the mathematics behind them — so that PRC could review and validate the approach, which it has done.

This part is limited to that combination step. It does not describe how PRC develops its underlying forecasts — PRC's own methodology governs that — and it does not describe how PPS staff decide which scenarios to propose in the first place, which is a separate planning and community engagement process.

In Brief: The Method in Eight Steps

- Start from PRC's forecast exactly as provided. This baseline is never altered.
- If a scenario closes a school outright, first adjust the historical residency table so that the closing school's attendance boundary is carried forward to the school or schools that absorb it.
- Once a proposed change takes effect in a given year, assume it continues in every later year of the ten-year forecast, unless a further change supersedes it.

- Decide which students are affected: either everyone in the specified grades right away, or a gradual, grade-by-grade phase-in that protects students who are already enrolled.
- Determine how many students each rule moves. A percentage below 100 moves that share of PRC's original projection; a percentage of exactly 100 moves everyone present, including students an earlier rule has just moved in.
- Move the affected students: either directly to one receiving school, or — when a school or program closes outright — split proportionally across the surrounding attendance boundaries, based on where those students have historically lived.
- Convert every fractional count leaving a given school, program, grade and year into whole numbers together, in a single step, so that rounding never inflates or reduces the district-wide total.
- Record the result as additions and subtractions layered on top of PRC's untouched baseline, then verify the finished scenario automatically before any figure is published.

The rest of this part walks through each of these steps in detail.

Key Terms

Term	Meaning
Baseline ("Status Quo")	PRC's original enrollment forecast, exactly as provided, before any scenario is applied.
Scenario	A proposed change PPS wants to model — for example, closing a program, merging two schools, or shifting a boundary — defined by which students are affected, where they would go, and when the change begins.
Rule	One line of a scenario. A scenario is normally made up of several rules, each covering one originating school, program category and grade range.
Cell	One school, one program category, one grade, one year. This is the smallest unit the model works in, and the unit at which rounding is settled.
Share rule	A rule whose percentage is below 100. It moves that share of PRC's original projection for the cell (Section 3.5).
Sweep rule	A rule whose percentage is exactly 100. It moves every student present in the cell at that point, including any who arrived from an earlier rule (Section 3.5).
Cohort	A group of students who entered school in the same year and move up one grade together each year, used to track a phased-in change as it follows the same students over time.

Term	Meaning
Ledger	The running record of every increase and decrease produced by a scenario, kept separately from — and without altering — PRC's original baseline forecast.
Successor boundary	The school or schools that absorb a closing school's attendance boundary, and therefore inherit its share of historical residency (Section 3.3).
Functional Capacity	The number of students a school building can reasonably serve, used together with modeled enrollment to calculate utilization.

2. The Three Inputs to the Model

2.1 PRC's Enrollment Forecast (the Baseline)

For every school, enrollment category, grade, and year, PRC supplies a single projected headcount. This complete set of numbers is the starting point — the "baseline" or "status quo" — against which every scenario is compared, and it is never edited or overwritten anywhere in the process.

PRC organizes its forecast into a small number of enrollment categories that distinguish why a student is enrolled where they are:

- **Residents Attending Neighborhood Program** — students who live within a school's official attendance boundary and attend that school's standard program. This is the core population most directly driven by neighborhood demographic change, and generally the largest share of the district's enrollment.
- **Non-residents Attending Neighborhood Program** — students attending a standard program outside their own attendance boundary, typically because of a grandfathered boundary change, an approved hardship transfer, admission by lottery to a focus option housed at a neighborhood school, or a family that moved but chose to let a student finish out their remaining grades at their original school.
- **Special Programs** — district-managed programs that are not tied to a residential boundary, including alternative and charter-style programs.
- **Language immersion programs** (for example, Spanish, Mandarin, Russian, Japanese, and Vietnamese Immersion) — these draw from two sources at once: neighborhood residents who choose the immersion track at their local school, and transfer students who travel from elsewhere in the district specifically for that language pathway.

PPS's modeling tool uses these same categories throughout, without reinterpreting them. A group of students keeps its original category before and after a scenario is applied, unless a scenario specifically calls for recategorizing them — which happens, for example, when a school closes and its "non-resident" population is reassigned into the standard resident category at their new home school.

2.2 Scenario Definitions

PPS planning staff describe each proposed change — a consolidation, a boundary shift, a program merger — in a simple, structured way. For every rule within a scenario, they specify:

- The originating school, program category, and grade range the change applies to;
- The receiving school and program category — or, for changes that dissolve a program or close a school outright, an instruction to return the affected students to their home attendance boundaries rather than to one specific receiving school. This option is used for students whose enrollment is not tied to where they live — non-resident, language immersion and special-program students — for the share assumed not to follow when their school or program closes or moves, or when a school stops enrolling non-residents. Students who live within a closing school's own boundary are instead sent to the named receiving school or schools, as in the worked example in Section 3.9;
- The year the change takes effect;
- What share of the projected group is affected, from a small fraction up to the entire group;
- Whether the change should apply to everyone in the affected grades immediately, or phase in gradually (Section 3.4 explains the difference); and
- A short written note describing what the rule is part of, in the planner's own words. This note is carried through the calculation and appears alongside every figure the rule produces, so that a reader of the detailed output can see the planning intent behind a number without consulting a separate document.

A single scenario is typically made up of many such rules — one for each school, program, and grade range it touches — which the modeling tool applies together, as described below. The percentage on each rule is always applied to a specific number. A percentage below 100 is applied to the PRC's original projection for the group; a percentage of exactly 100 percent instead moves every student in the group, including any moved there by rules the tool applies before it. Section 3.5 explains this distinction and the order in which the rules are applied.

2.3 Historical Residency Proportions

When a scenario calls for returning students to their home attendance boundaries rather than sending them to one specific receiving school — for example, non-resident, immersion or special-program students whose school or program closes (Section 2.2) — the tool needs to know how a program's enrolled population has actually been distributed across the surrounding boundary areas. PPS maintains this as a supporting table: for a given school, program, and grade band, the share of enrolled students who have, on average over the three most recent school years, lived in each nearby attendance boundary.

For any given school, program, and grade band, these shares always add up to 100 percent across every boundary area that program has historically drawn from. Some of that draw is legitimately from outside the district; where that is the case it is carried as such, and students apportioned there are understood to be leaving PPS rather than arriving at another PPS school.

This table describes history, and history does not know which schools a scenario is about to close. Section 3.3 describes the adjustment the tool makes before each scenario runs so that a closing school's boundary

is carried forward correctly. The table is used only for the boundary-based redistribution mechanism described in Section 3.6, and has no effect on scenarios that specify a direct, one-to-one destination school.

3. How a Scenario Is Applied, Step by Step

3.1 Establishing the Baseline

Before any scenario is considered, the tool takes PRC's complete forecast and sets it aside, untouched, as the reference point every scenario will be compared against. Every subsequent calculation adds to or subtracts from this reference — it never changes the reference numbers themselves. PRC's original forecast, unaltered, therefore remains available for inspection at every stage, alongside however many alternative scenarios PPS chooses to evaluate.

3.2 A Change, Once Made, Carries Forward

A scenario rule takes effect starting in a specified year and, once in effect, is assumed to remain in place for every later year of the ten-year forecast, unless a separate rule specifically supersedes it. In other words, a scenario describes a new steady state going forward, not a one-time event confined to a single year: a boundary change made effective in the fourth year of the model is assumed to still be in place in the tenth year, consistent with how these decisions function in practice. This lets PPS see a full ten-year trajectory for every school under a proposed change, not just a single snapshot year.

3.3 Preparing the Residency Table When a School Closes

The historical residency table described in Section 2.3 records where a program's students have lived. If a scenario closes a school, that school's attendance boundary still appears in the table as a place students live — and without an adjustment, a redistribution elsewhere in the district could assign students to a school the same scenario has just closed.

The tool therefore prepares a corrected copy of the residency table before each scenario runs. The master table is never edited; each scenario works from its own corrected copy.

Identifying a closure. A school is treated as closed in a given year when every group and grade it is projected to hold that year is fully reassigned elsewhere. A single program leaving is not a closure: a scenario can move an immersion program from School A to School B while School A continues to operate its neighborhood program, and School A is correctly not treated as closed.

Identifying the successors. A closing school's attendance boundary does not disappear — the families in it are absorbed by whichever school or schools the scenario reassigns its resident students to. The tool reads those destinations, and their relative shares, from the scenario's own rules, so the boundary follows the students rather than being assigned by a separate assumption.

Carrying the boundary forward. Every reference to the closing school as a place students live is replaced by one entry per successor, each carrying the original share multiplied by that successor's share of the reassignment. Suppose School A closes, with its resident students divided evenly between School B and School C, and suppose School A's boundary historically accounted for 8 percent of the students in a program at School D. In the corrected table that 8 percent is carried as 4 percent living in School B's

boundary and 4 percent in School C's. The families have not moved; the boundary they live in now belongs to two different schools, and the total is unchanged.

Where a successor is itself closed by the same scenario, the tool follows the chain through to the schools that remain open, composing the shares along the way. Where the successors differ by grade — for example, elementary grades to one school and middle grades to another — the boundary is carried forward separately for each grade range. Where a scenario closes a program that has no residential boundary at all, there is nothing to carry forward and nothing is changed.

Every reassignment the tool makes is written to a "Closed School Successors" report alongside the results, naming each closed school, the year it closes, each successor, and the share carried to it.

3.4 Determining Which Students Are Affected: Immediate vs. Phased-In Change

Not every change is meant to apply to every enrolled student the moment it takes effect. PPS planning staff mark each rule as either immediate, or — by leaving this unspecified, which is the current default — a gradual, phased-in change.

Immediate: every student in the specified grade range at the originating school and program is affected, starting in the effective year and in every year after.

Phased (default): the change protects students already enrolled above the starting grade, applying instead only to the incoming class at the youngest specified grade, and to every class that enters afterward. As those students move up a grade each year, the change follows them — the same idea behind standard cohort-survival forecasting, where a group of students is tracked not by name but by the size of the class that entered together and ages up as a group. Practically, this means a change phases in one grade at a time: in the first year, only the newly entering class at the starting grade is affected; the next year, that same class — now one grade higher — is still affected, and a new entering class is added as well; and so on, until the entire specified grade range is fully phased in and remains so for every year after.

	Immediate	**Phased (default)**
Students affected in the first year	Everyone in the specified grade range	Only the incoming class at the youngest specified grade
Students affected in later years	Everyone in the specified grade range, every year	That same class, plus each new incoming class, until the full range is phased in
Students already enrolled above the starting grade when the change begins	Included right away	Grandfathered — they finish their remaining grades at the original school

Framed around real families: under a phased change, a student already partway through the affected grade range when the change is announced finishes out their remaining grades at their original school; a student entering the starting grade in the year the change begins — or in any year after — is affected for the entire specified grade range.

3.5 What a Percentage Means: Share Rules and Sweep Rules

A scenario rule specifies what share of a group is affected. Because a single school, program and grade can be touched by several rules at once — some students moving to one receiving school, some to another, some returning to their home boundaries — the tool must be precise about what each percentage is measured against. It distinguishes two cases.

A share rule is any rule below 100 percent. It moves that share of PRC's original projection for the school, program, grade and year — never a share of whatever an earlier rule happened to leave behind. Every share rule drawing on the same group is therefore measured against the same number, so complementary shares add up exactly as written.

This is what allows a scenario to divide a group cleanly. If a group of 100 projected students is split by three rules — 50 percent returning to home boundaries, 25 percent to School B, and 25 percent to School C — all three are measured against the same 100, they remove 50, 25 and 25, and the originating group empties completely. Had each rule instead been applied to the students remaining after the previous one, the same three rules would remove 50, then 12.5, then about 9.4, leaving more than a quarter of the students enrolled at a school the scenario closes.

A sweep rule is a rule of exactly 100 percent. It moves every student present in that group at that point, which includes any students an earlier rule in the same scenario has just moved in. This is deliberate: it is what allows a closing school to land on exactly zero even when the scenario routes students through it. A school projected to have 31 students that also receives 1 student from another rule will show a sweep of 32. Rules below 100 percent are applied first, then 100 percent rules that return students to their home boundaries, then 100 percent rules to named schools; each moves the students present after the stages before it.

The distinction concerns the denominator, not the destination. Both kinds of rule may send students directly to a receiving school or return them to their home boundaries, as described in Section 3.6.

In practice this means a closure should be written either as a single 100 percent rule, or as a set of share rules that add to 100 percent. One case deserves note: if a school's closure is written entirely as share rules totaling 100 percent, and that same school also receives students from another rule in the same scenario, the share rules will not carry those arrivals away, because they are measured against PRC's projection rather than against the arrivals. For example, if School B is projected to have 40 students in a grade and its closure is written as 60 percent to School C and 40 percent to School D, those two rules move exactly 40 students. If another rule in the same scenario also sends 2 students from School A to School B in that grade, those students are left behind at a school that has closed.

This case is narrow. It arises only when a scenario sends students into a school that the same scenario also closes. One way students could arrive at a closing school — a closed program returning students who live in its boundary — cannot cause it, because the adjustment in Section 3.3 sends those students to the schools that absorb the boundary instead. What remains is a scenario in which one school's students are sent to a second school that itself is closing. The reliable way to avoid it is send the first school's students directly to the final destinations. Closing the second school with a single 100 percent rule also works when the students arrive by a rule below 100 percent. In the scenarios PPS has evaluated to date, the verification step has not found this case.

If it did occur, it does not pass unnoticed. The verification step described in Section 5 reports the students left behind as an error, naming the school, program, grade and year, and they are resolved as described there; the scenario's rules are corrected and the scenario is run again, rather than the students being moved by hand.

3.6 Two Kinds of Destinations: Direct Transfers and Boundary-Based Redistribution

Once the tool has determined how many students a rule moves, it sends them in one of two ways, depending on how the rule is written.

Direct transfers are the more common case: the affected students in a specific school, program, and grade range are reassigned to one specific receiving school and program. If a scenario specifies moving 100 percent of Kindergarten through Grade 5 from School A to School B starting in a given year, then every projected student in that group — at School A that year, and in every following year's incoming Kindergarten class — is reassigned to School B for as long as the rule remains in effect.

Boundary-based redistribution applies when a scenario instead dissolves a program or closes a school and returns its students to their home attendance boundaries, rather than sending them to one specific receiving school. Because those students could live across several different surrounding boundary areas, the tool cannot send them to just one place. Instead, it uses the historical residency proportions described in Section 2.3 — as corrected for any school the scenario closes, per Section 3.3 — to split the departing group across every boundary area it has historically drawn from, in the same shares history shows. For example, if a program with a historical draw of 50 percent from Boundary Area 1, 31 percent from Boundary Area 2, and 19 percent from Boundary Area 3 is discontinued, a departing group of students from that program is split across the three areas in those same proportions, each added into that area's standard neighborhood program count.

In both cases, the number of students affected and the year the change takes effect come from the rule itself; only the destination logic differs.

Grade-level detail within a redistribution

The residency shares described in Section 2.3 are quoted for a grade band — typically elementary and middle grades — rather than for a single grade. The same share therefore drives every grade in the band, and the model has no grade-level signal within it. This has one consequence worth stating plainly and one worth guarding against.

The consequence worth stating is that the grade a returning student is assigned to should not be read as a grade-level forecast. The model knows how many students each boundary area should receive across the band, but not which grades they are in. For example, suppose a closing program sends 25 students home from each grade, Kindergarten through Grade 5, and one boundary area has historically supplied 2 percent of the program's students. That area's exact share is half a student in every grade, or 3 students across the band. Because the model reports whole students, the area's students appear in some grades and not others. The band total is a meaningful projection; which grades show a student is not, because every grade has the same claim of half a student and the data gives no reason to prefer one grade over another. Part 3 walks through a real case of this kind.

The same caution applies when comparing scenarios. Two scenarios that make the same change can place a returning student in different grades, because each grade is settled together with every other boundary

area in the band, and a closure elsewhere in one scenario changes those other shares. Compare scenarios on band totals for returning students, not grade by grade.

Computing the shares grade by grade instead would not improve this. A program of roughly 25 students per grade drawn from 40 or more boundary areas produces per-grade, per-boundary counts that are almost entirely zeros and ones, and a three-year average of those figures is noise rather than signal. For example, if one boundary area happened to send the program a first grader in each of the last three years but no second graders, grade-level shares would credit it with 4 percent of first grade and none of second grade — a difference created by which families happened to enroll in those particular years, not a pattern there is any reason to expect to continue. The band-level average is the appropriate granularity for the data available.

The risk worth guarding against is subtler. Because the same share applies to every grade in the band, settling each grade's split independently would repeat one near-identical rounding decision once per grade, always in the same direction. A boundary whose exact claim is slightly more than half a student in each grade would collect a whole student in every grade of the band; one whose claim is slightly less would collect none in any of them. Over a six-grade band that turns an exact claim of, for example, 3.6 students into 6, or 2.4 into none — in the same direction in every forecast year, since the shares do not change. The tool avoids this by settling the band as a whole, as described in Section 3.7, which works through this example.

3.7 Keeping Every Student Whole: The Rounding Problem and How It's Solved

Applying a percentage to a whole-number forecast will usually produce a fraction. Thirty percent of seventeen students is 5.1 — and while a forecast can describe a fraction of a student on paper, a modeled outcome cannot. This is not a new problem; it is a version of a well-known issue in applied mathematics, the same one that arises whenever a fixed total has to be divided among several groups in proportion to unequal shares, such as apportioning a fixed number of legislative seats among states of different populations.

Rounding each share independently — the obvious first approach — can quietly gain or lose whole students. If ten departing students are split three ways in roughly equal shares (3.33, 3.33 and 3.33), rounding each independently gives 3, 3 and 3 — only nine students, with a tenth vanishing from the model even though ten clearly left the originating school.

Two decisions follow from this: which numbers must be rounded together, and how the rounding itself is done.

One rounding decision per cell

All students leaving one school, program, grade and year are converted to whole numbers together, in a single step — regardless of how many rules are moving them or whether those rules send students to different receiving schools or back to home boundaries. The cell, not the individual rule, is the unit at which rounding is settled.

Settling each rule separately would break the total. Suppose 27 students in one grade are divided evenly between two receiving schools. Each rule's exact share is 13.5. Rounded independently, both round the same way, and the two together remove either 26 or 28 students from a group of 27 — leaving a student

stranded at a school the scenario has closed, or removing one who was never there. Settled together, the pair is required to total exactly 27, and one school receives 14 while the other receives 13.

The same principle governs a boundary-based redistribution, one grade band at a time. Because that step builds on the rounding method described next, it is explained with a worked example, under “Setting a grade band as a whole” at the end of this section.

How the whole numbers are chosen

The tool uses the Largest Remainder Method, also known as Hamilton’s method in the apportionment literature. Each site first receives the whole-number part of its exact share. That will normally leave a small number of whole students still to be assigned, and those go, one each, to the sites whose fractional remainders are largest.

Stated formally: for a group of n sites sharing a departing or arriving student count, let v_i be the exact fractional share calculated for site i. Write v_i as a whole-number part a_i plus a remainder r_i between 0 and 1, and let k be the number of whole students left over once every site has received its a_i:

$k = \text{round}(v_1 + v_2 + \ldots + v_n) - (a_1 + a_2 + \ldots + a_n)$

Each site receives a_i. The k sites with the largest remainders each receive one additional student.

This is not an approximation. Ranking the remainders from largest to smallest as $r_{(1)}$ through $r_{(n)}$, the largest gap between any site’s exact share and the whole number it receives is $\max(1 - r_{(n)}, r_{(n+1)})$, and no other assignment can do better: any alternative must round up a site with a remainder no larger than $r_{(n)}$, producing a gap of at least $1 - r_{(n)}$, and must round down a site with a remainder no smaller than $r_{(n+1)}$, producing a gap of at least $r_{(n+1)}$. The same assignment also rounds up the largest total remainder available, so it minimizes the sum of all the gaps as well as the largest one.

This reaches the same optimum described by the constrained-optimization formulation in the earlier draft of this methodology — the smallest possible largest gap, subject to the whole numbers summing to the correct total. It reaches it directly rather than by search, which means the tool no longer depends on a third-party optimization solver, runs substantially faster, and produces identical results on any computer and in any future run.

When two sites tie

Two sites can have identical remainders at the point where the leftover students run out. The clearest case is an odd number of students split evenly two ways: each side’s exact share ends in one half, one side must receive the extra student, and neither has a better claim to it. Every such combination scores identically on both measures above, so the choice is genuinely free — but it should not always fall the same way.

The tool settles these ties by a fixed rule that depends on the cell being calculated and on the names of the sites involved. Two consequences follow. Within a grade range, the extra student alternates between the receiving schools rather than accumulating at one of them: under the previous approach, a school could receive the odd student in every grade of a K–5 span, ending several students ahead purely as a by-product of how ties were resolved. And because the rule depends only on the inputs, it produces the same answer on every computer and in every rerun.

The tie-breaking rule is governed by a published setting recorded with the model. Re-running a scenario under a different setting produces a different, equally valid resolution of the same ties, which allows PPS to confirm directly that no conclusion depends on how ties happened to fall. In testing to date, changing this setting has moved no school by more than three students in any year and has never changed a school's closure, a district total, or a utilization conclusion.

Direct transfers need no second rounding

For a direct transfer with a single receiving school, no second rounding decision is needed at all: whatever whole number of students is confirmed as leaving the originating school is credited to the receiving school as that same exact number. Direct transfers are therefore conserved exactly, down to the student. The Largest Remainder Method is applied at two points only: when settling the departures from a cell, and when splitting a confirmed whole-number departure across the boundary areas it returns to.

Settling a grade band as a whole

When students return to their home boundaries, every grade in a band is split using the same residency shares, so settling each grade on its own would repeat the same rounding decision in every grade, always in the same direction (Section 3.6). The tool therefore settles the band as a whole. It still works through the band one grade at a time, and every grade still places exactly the students who left it. What changes is the amount each boundary area is treated as entitled to in that grade: not its share of that one grade alone, but what it is still owed across the band — its share of every student who has left the band so far, less the students it has already received. Within each grade, the whole numbers are then chosen by the Largest Remainder Method, exactly as described under "How the whole numbers are chosen" above.

An example makes this concrete. Suppose a program closes and sends 2 students home from each grade, Kindergarten through Grade 5, and that its students have historically lived in three boundary areas: 50 percent in Area 1, 30 percent in Area 2 and 20 percent in Area 3. In each grade, the exact shares are 1.0, 0.6 and 0.4 students; across the band, they are 6.0, 3.6 and 2.4. The two approaches compare as follows:

	Area 1	Area 2	Area 3
Exact share, Kindergarten–Grade 5	6	3.6	2.4
Each grade settled on its own	6	6	0
Band settled as a whole	6	4	2

Settled one grade at a time, every grade comes out the same. Area 1 receives its whole student, and the second student goes to Area 2, whose remainder of 0.6 is larger than Area 3's 0.4. Repeated in all six grades, that gives Area 2 six students and Area 3 none, even though one in five of the program's students has historically lived in Area 3.

Settled as a band, Area 1 still receives one student in every grade, since it is owed exactly one each time. The second student goes to whichever of the other two areas is owed more at that point:

Grade	Still owed to Area 2	Still owed to Area 3	Second student goes to
Kindergarten	0.6	0.4	Area 2
Grade 1	0.2	0.8	Area 3
Grade 2	0.8	0.2	Area 2
Grade 3	0.4	0.6	Area 3
Grade 4	1	0	Area 2
Grade 5	0.6	0.4	Area 2

Take Grade 1. By then 4 students have left the band, so Area 2's share of them is 1.2; it received 1 in Kindergarten, so it is owed 0.2. Area 3's share is 0.8 and it has received no one, so it is owed 0.8, and the student goes to Area 3. By the end of the band, Area 2 has 4 students against an exact share of 3.6 and Area 3 has 2 against 2.4 — each within 0.4 of a student, where settling grade by grade missed each by 2.4.

Nothing already placed is ever changed. When an area receives more than its running share in one grade — as Area 2 does in Kindergarten, receiving 1 student against a share of 0.6 — that student stays where they were placed. The extra simply reduces what the area is owed in the grades that follow, which is why Area 2 receives no one in Grade 1, and an area that is further ahead receives no one until its share catches up. The adjustment only ever works forward; once a grade has been settled, it is not revisited.

In practice the tool does not always begin at Kindergarten. Each year it starts the band at a different grade, chosen by a fixed published setting, so that the rounding does not fall in the same grade in every year of the forecast; Part 3 explains why that matters for how the output reads. In testing against a full ten-year scenario, settling the band as a whole reduced the average error in a receiving school's band total from about 1.4 students to about 0.2, and eliminated every case where a school's band total was wrong by two or more students.

3.8 Recording the Result: An Audit Ledger, Not an Overwrite

Rather than editing PRC's baseline numbers directly, the tool records every scenario's effect as a running ledger of paired entries, much like an accounting ledger: a decrease at the originating school, program, grade, and year, and a matching increase at the receiving school, program, grade, and year — always adding to exactly the whole-number amount determined in Section 3.7. The baseline itself is never edited.

This has several practical benefits. PRC's original forecast remains fully intact and inspectable at every stage, as the "Status Quo" reference point. Every change traces directly back to the specific scenario rule that produced it. Multiple scenarios can be modeled side by side against the same untouched baseline without interfering with one another or with PRC's source data. And a school's fully modeled enrollment, for a given scenario, year, and grade, is simply its baseline count plus every ledger entry that applies to that same cell under that scenario.

Each ledger entry also records the reasoning behind its own figure: which rule produced it, what the rule's percentage was applied to, the exact fractional result before rounding, whether the figure was rounded up or down and why, and the planner's written note for the rule. Section 6 describes how to read this record.

3.9 Worked Example (Illustrative Only)

The example below is entirely generic and does not describe any real school, program, or projected count; it is meant only to show how the pieces above work together.

Suppose School A is proposed for closure starting in a given year. Its resident students are divided evenly between School B and School C. Its non-resident students — those attending School A from outside its boundary — are treated differently: half are assumed to return to the standard program at their own home school, and the remaining half follow the residents, one quarter to School B and one quarter to School C. The change is marked as immediate.

Follow one grade of the non-resident group, for which PRC projects 18 students in the first effective year.

- **Preparing the residency table.** Because School A closes entirely, its attendance boundary is carried forward before anything else happens (Section 3.3). School A's residents split evenly between School B and School C, so every place the residency table records a student as living in School A's boundary is replaced by two entries at half the share, one for School B and one for School C.
- **Determining the amounts.** All three rules affecting this group are share rules, so all three are measured against PRC's projection of 18 (Section 3.5). They remove 9.00, 4.50 and 4.50 students respectively — exactly 18 in total, so the group empties completely.
- **Converting to whole students.** The three fractional departures are settled together (Section 3.7). They must total 18. The first is already whole at 9. The other two are tied at 4.50, and one of them must give up the extra student; the tie-breaking rule assigns it, and the departures are recorded as 9, 5 and 4. In the next grade up, where the projected count differs, the tie may well fall the other way.
- **Distributing the boundary return.** The 9 students returning to their home boundaries are split using the corrected residency table (Section 3.6). If that table shows shares of 50, 31 and 19 percent across three boundary areas, the exact shares are 4.50, 2.79 and 1.71. Each area first receives its whole-number part — 4, 2 and 1, totaling 7 — which leaves 2 students still to assign, and those go to the two largest remainders, 0.79 and 0.71. The tool therefore assigns 4, 3 and 2, totaling exactly the 9 who left. Note that the first area is not rounded up even though its exact share ends in one half: rounding each figure independently to the nearest whole number would give 5, 3 and 2 — ten students from a departure of nine. Settling the three together instead holds the total at nine and keeps the largest single gap to 0.50 of a student, which is the smallest it can be.
- **Recording the result.** The ledger records three decreases against School A for this group, grade and year — 9, 5 and 4 — and matching increases: 5 at School B, 4 at School C, and 4, 3 and 2 at the three boundary areas' standard neighborhood programs. PRC's original forecast for School A is left completely untouched.

- **Reading the result.** A planner asking what School A's modeled enrollment is for this group and grade would see PRC's original 18 plus decreases totaling 18, netting to zero, consistent with the school having closed. A planner asking about School B would see PRC's original forecast plus its share of the transfer and its share of the returning students — including students who live in what used to be School A's boundary.

4. From Modeled Enrollment to Building Utilization

Once a scenario's fully modeled enrollment is assembled for every school, grade, and year, PPS compares it against each building's functional capacity – the number of students a building can reasonably serve – to calculate a utilization rate (modeled enrollment divided by functional capacity) and the number of seats available or short at each site.. This comparison is produced twice for every school and year: once using PRC's forecast before any scenario changes, and again using the fully modeled result after a scenario's changes are applied, so that planners can see directly how a proposed change would affect each building's utilization.

5. Built-In Verification and Reconciliation

Every scenario is checked automatically after it is calculated and before any figure is published. The checks are deliberately mechanical: each one states a property the finished numbers must have if the method has been applied correctly, and reports every instance where they do not.

- Every school, program and grade that a scenario reassigns in full ends at exactly zero. A residual of even one student is reported as an error, identified down to the grade and year. The students are not moved by hand; instead, the scenario's rules are corrected and the scenario is run again, so that every student who moves does so through a rule recorded in the ledger (Section 3.8), Section 3.5 describes one way a residual can arise and how it is avoided.
- No school's modeled enrollment is negative in any group, grade or year.
- District-wide totals are unchanged year by year: the fully modeled result must reconcile exactly to PRC's baseline total, since a scenario moves students between schools but never creates or removes them. Students apportioned to an out-of-district boundary (Section 2.3) are carried on their own line, so this reconciliation still holds; the school-by-school tables leave that line out, so their sum is below PRC's total by exactly those students.
- No students are routed into a school the scenario closes, in the closure year or any year after.
- No set of rules claims more than 100 percent of the same group, which would otherwise drive a school's enrollment below zero.
- Every receiving boundary is a school that exists in PRC's forecast. Out-of-district destinations are recognized as legitimate and reported separately from genuine errors.

Findings are written to a "Guard Findings" report alongside the results, giving the scenario, school, group, grade, year, the check that failed, and a plain description of the discrepancy. Findings are graded: an error means the numbers should not be published as they stand, while a warning marks something that may be

defensible but deserves a human decision. The tool can be set to halt the run outright if any error is present, so that a scenario with a known defect cannot reach a published utilization table.

Conditions that could previously have passed unnoticed now stop and report. A redistribution with no matching residency record, a boundary area left with no successors after a closure, and a group of students that cannot be placed are each named explicitly rather than being silently dropped.

Alongside the findings, the tool publishes the "Closed School Successors" report described in Section 3.3, so that every adjustment made to the residency table is visible next to the results it produced.

6. How to Read the "rounded" Tab

The "rounded" tab is the ledger described in Section 3.8, published in full. Every row is either a line of PRC's baseline or a single increase or decrease produced by one scenario rule. Each row is self-contained: it states which rule moved the students, what the rule's percentage was applied to, where they went, and how the figure became a whole number. No lookups into other tabs are required to check a number.

The tab is a record of the calculation, not a summary of it. A school's modeled enrollment for a given scenario, year, group and grade is the sum of the baseline row and every ledger row for that same cell — which is how the summary and utilization tabs are built.

6.1 Column Reference

Column	What it tells you	Defined in
school, group, grade, year	The cell this row affects. For a decrease this is the originating school; for an increase, the receiving school.	Section 2.1
count	The students this row adds or removes. Decreases are negative. A departure can show 0 when its fraction rounded away; it then has no matching arrival.	Section 3.8
direction	Baseline, Departure, or Arrival.	Section 3.8
from_school, from_group	Where the students are leaving.	Section 3.6
to_school, to_group	Where they are going. On a departure that disperses students, this reads "home boundaries"; the individual receiving boundaries appear on the matching arrival rows.	Section 3.6
rule_kind	Share or Sweep — what the rule's percentage was measured against.	Section 3.5
rule_pct	The percentage as written in the scenario definition.	Section 2.2

Column	What it tells you	Defined in
basis_count	The number that percentage was applied to: PRC's projection for a share rule, the students present for a sweep.	Section 3.5
boundary_share	This boundary's historical share of the originating program, on redistribution arrivals only.	Section 2.3
unrounded_count	The exact fractional result before rounding. Negative on departures, like count.	Section 3.7
rounded_count	The whole number assigned. Negative on departures, like count.	Section 3.7
rounding	Up, Down, or Exact — stated in students, so "Up" always means one more student moved.	Section 3.7
explanation	The row restated as one sentence, for readers who will not pivot the data.	—
calc_note	Populated only when something happened that the other columns do not make obvious. Blank on ordinary rows.	Section 6.3
scenario_note	The planner's written note for the rule that produced this row.	Section 2.2
cell_id	A single filter key for the cell this row affects: one school, group, grade and year. For a departure, this is also where rounding was settled.	Section 3.7
scenario_name, audit_log, transaction_id	Which scenario (reads "Baseline" on PRC's baseline rows), which rule, and the key linking a departure to the arrivals it produced: several for a return to home boundaries, none when the departure rounded to zero.	Section 3.8
eval_scenario	The scenario run that this row belongs to. PRC's baseline rows repeat in every run with scenario_name "Baseline", so filter on eval_scenario, not scenario_name, to see one scenario's complete picture. |	Section 3.8

6.2 Reading One Cell End to End

The clearest way to read the tab is to filter on a single cell_id, which brings together the baseline and every rule that touched it. Taking the non-resident group from the worked example in Section 3.9 — School A, one grade, the first effective year — the tab shows four rows:

Row	What the tab says
18	Baseline. PRC projection.
−9	Departure to home boundaries. Share rule, 50% of basis 18 = 9.00. Rounding: Exact. Explanation: 50% of the 18 students PRC projects for this cell = 9.00, moving to home boundaries.
−5	Departure to School B. Share rule, 25% of basis 18 = 4.50. Rounding: Up. Explanation: 25% of the 18 students PRC projects for this cell = 4.50, moving to School B. Rounded up to 5 so this cell's departures total exactly 18. Note: tied at 0.50 of a student with School C; the tie-break rule awarded the extra student here.
−4	Departure to School C. Share rule, 25% of basis 18 = 4.50. Rounding: Down. Explanation: 25% of the 18 students PRC projects for this cell = 4.50, moving to School C. Rounded down to 4 so this cell's departures total exactly 18. Note: tied at 0.50 of a student with School B; the extra student went to School B.

Read together, the four rows account for the cell completely: 18 projected, 18 removed, ending at zero as a closure requires. The two rules that appear identical in the scenario definition — both 25 percent, both producing 4.50 — are distinguished by their receiving school, and the reason one became 5 and the other 4 is stated on the rows themselves.

The matching arrival rows appear at the receiving schools and boundary areas. A redistribution arrival carries the boundary share that sent the students there, and reads, for example: 3 of the 9 students returning from School A grade 3 to their home boundaries. Boundary Area 2 is 31.0% of School A's historical residency for this group and grade = 2.79. Rounded up to 3.

6.3 Checking a Figure by Hand

Any number in the tab can be verified in three steps, each corresponding to a section above.

- **The rule's arithmetic (Section 3.5).** Multiply rule_pct by basis_count. The result should equal unrounded_count, which is negative on a departure because students are leaving. If rule_kind is Sweep, rule_pct is 100 percent, so unrounded_count simply equals basis_count. This check works on every departure, and on every arrival from a direct transfer, which carries the same rule_pct and basis_count as its departure. An arrival from a redistribution follows different arithmetic; check it with the third step below.
- **The rounding (Section 3.7).** Filter on cell_id and sum count across the rows where direction is Departure. That total should be a whole number equal to the sum of the unrounded departures, rounded once. The rounding column shows which rows absorbed the fraction.
- **The redistribution (Section 3.6).** On a redistribution arrival, multiply boundary_share by the total number of students who left that cell for their home boundaries. The result should

equal unrounded_count, and the shares across all arrivals from that departure should sum to 100 percent.

Filtering calc_note to non-blank rows shows everything unusual the run did. Four notes are possible, and a row can carry more than one:

Note	Meaning
A tie	Two rules or two boundaries had identical claims on one student. The note names the other party and states which way the tie fell (Section 3.7).
A reassigned boundary	The receiving school's historical share includes the attendance boundary of a school this scenario closes. The note names the closed school and the share carried over (Section 3.3).
A sweep that collected arrivals	A 100 percent rule moved more students than PRC projected for that cell, because students had arrived from an earlier rule. The note gives the number that arrived (Section 3.5).
A phased rule	The rule applies only to cohorts that have reached the starting grade since the year of change, so it does not affect every grade in its stated range in early years (Section 3.4).

One presentational note: the numeric columns are written as zero rather than left blank on baseline rows, where a rule percentage and basis do not apply. The direction column reads "Baseline" and the explanation reads "PRC projection" on every such row.

7. Summary

Throughout this process, PPS's modeling tool is built around five guarantees. PRC's original forecast is preserved exactly and is never overwritten. Every scenario's effects are layered on top of that forecast as a fully traceable set of additions and subtractions, each carrying the reasoning behind its own figure. Percentage-based moves are measured against PRC's projection and converted to whole numbers a cell at a time, so that complementary rules add up as written and rounding never gains or loses students in aggregate. Changes are modeled in a way that reflects how they actually unfold in practice — carrying forward year over year, protecting students already partway through an affected grade range by default, and carrying a closing school's attendance boundary forward to the schools that absorb it. And every finished scenario is verified automatically against a set of stated properties before any figure is published.

This methodology has been reviewed by the PRC. PPS welcomes questions about the approach described above.

Part 3. Reading Grade-Level Detail in Redistribution Results

Why students returning to a neighborhood school do not appear to move up a grade

A companion to Part 2 · August 2026, revised October 2026

About this part: This is a supplement to Part 2, written because one feature of the model's output reliably surprises people who look at it closely — including the staff who built it. It concerns how to read grade-by-grade figures when a school or program closes and its students return to their home attendance boundaries. Nothing here describes a defect. It describes a limit on what the underlying data can support, and the choice PPS made about how to present a result honestly within that limit.

A note on the figures: The numbers below are taken from an actual run of an earlier exploratory scenario, not from the scenarios currently proposed; school names have been replaced with generic labels so that this part can be shared independently of any particular proposal. "Program P" stands for a special program proposed for closure; "School R" stands for one neighborhood school whose attendance boundary supplied a small share of that program's students.

1. The observation

When a special program closes and its students are returned to their home attendance boundaries, a reader looking at a single receiving school's results year by year may notice something odd. The students appear in the same grade in every year of the ten-year forecast.

In the case that prompted this part, School R gained two students in second grade in the first year of the closure — and two students in second grade in every year afterwards, for nine consecutive years, with nothing in any other grade until late in the forecast.

That looks wrong in a specific way. If a second grader returns to School R this year, common sense says they should be a third grader there next year. A stack of students sitting permanently in one grade looks like the model has failed to age anyone.

It has not. The explanation involves three separate things that are easy to run together, so this part takes them one at a time.

2. The short answer

The model does not contain students. It contains counts.

Nothing inside the model has an identity that could get a year older, so nothing can move from one grade to the next. Each year's figure is computed from scratch.

Cohort progression has already happened before the model runs.

PRC's forecast is built by tracking cohorts as they age. The model applies a percentage to numbers that already carry that work.

The repeated grade was a rounding artifact, not a statement about students.

The model's honest answer for School R is about half a student in every grade in every year. Whole students have to be printed somewhere, and they were all landing in the same row.

3. What the model is made of

The model works on counts held in boxes. A box is one school, one enrollment category, one grade, one year. PRC supplies a number for every box; a scenario adds to some boxes and subtracts from others.

When the tool "moves students," it subtracts a number from one box and adds the same number to another box in the same grade and the same year. There is no student record, no roster, no identifier attached to anyone. Nothing in the model is the kind of thing that could have a birthday.

So the question "why don't these students move up a grade next year?" has an unexpected answer: there is nothing there to move. The figure in School R's third-grade box for next year was not produced by aging this year's second-grade figure. It was calculated independently, from PRC's forecast for that school, that grade, and that year.

4. Where cohort progression actually happens

Cohort progression is central to this work — it simply happens earlier in the chain than people expect.

PRC builds its forecast using cohort survival: it tracks a group of students as they move up one grade each year and estimates how many remain. That work is finished before PPS receives the file. In the run behind this part, PRC projects that Program P has 26 students in second grade in one year and 27 in third grade the year after. Those are largely the same children, one year older. The progression is already inside those two numbers.

The model then does something much simpler. School R's attendance boundary historically supplied a small, fixed share of Program P's enrollment, so the tool multiplies that share by each year's projected count:

Year 1, grade 2: 26 students × School R's share = 0.44 students returning to School R

Year 2, grade 3: 27 students × the same share = roughly the same figure again (the table in Section 5 has every grade and year)

The progression lives in 26 becoming 27. That is PRC's work, and the multiplication rides on top of it.

A useful way to picture it is a pipe with a small tap. PRC decides how much water flows through the pipe each year at each grade — all the demographic and cohort reasoning happens there. The tap diverts a fixed fraction into School R. The tap does not need to know anything about cohorts, because the pipe already does. Building cohort logic into the tap as well would do the same work twice.

5. What the model actually knows about a receiving school

Here is the whole of what the model knows about School R under this scenario: its exact expected arrivals, before anything is rounded, for every grade in every year.

Grade	2027-2028	2028-2029	2029-2030	2030-2031	2031-2032	2032-2033	2033-2034	2034-2035	2035-2036
KG	0.41	0.44	0.42	0.41	0.41	0.41	0.41	0.41	0.41
1	0.44	0.44	0.47	0.45	0.44	0.44	0.44	0.44	0.44
2	0.44	0.44	0.44	0.46	0.44	0.44	0.44	0.44	0.44
3	0.43	0.44	0.44	0.44	0.47	0.45	0.45	0.45	0.45
4	0.40	0.45	0.45	0.46	0.46	0.49	0.47	0.46	0.46
5	0.52	0.40	0.45	0.45	0.46	0.46	0.49	0.47	0.46

Every one of these 54 figures falls between 0.40 and 0.52 of a student. It is a flat plane, not a ridge. There is no grade that stands out and no year that stands out.

This flatness is the signature of cohort progression working, not failing. If there were one identifiable group of returning students aging through the grades, this table would show a diagonal ridge — larger figures stepping down and to the right as that group moved up a grade each year. There is no such ridge, because the situation is a steady state. The program closes permanently, so in every year of the forecast a fresh set of students who would have attended it return home instead. Each year a new kindergarten cohort joins the flow and a fifth-grade cohort ages out of it, and the expectation settles at roughly the same value in every grade, every year.

Real children do of course progress. The students School R gains in the first year are largely the same people it has in the second year, one grade higher. But that continuity is invisible in the totals, because it is exactly offset by the cohorts entering and leaving. What remains is a plateau.

6. Why whole numbers force an arbitrary choice

A forecast can describe 0.44 of a student. A modeled result cannot. Every figure PPS publishes has to be a whole number of children.

School R's exact claim comes to roughly 2.6 students a year across the six elementary grades. The model must show a whole number, so it shows 2 in most years and 3 in others. That number then has to be

placed in one or two of the six grade rows — and the table in Section 5 gives no basis whatsoever for choosing which. Every row has an equal claim of about half a student.

Whichever rows are chosen, the result is wrong by roughly the same amount in every box: the model prints 1 where the truth is 0.44, and 0 where the truth is 0.44. This is unavoidable. It is not a shortcoming of the method but a consequence of reporting a fractional expectation as whole children.

What is avoidable is the shape of that error.

7. The original behavior, and what changed

The tool settles a whole grade band together, so that a receiving school's total across the band is accurate (see Part 2, Section 3.7). Originally it worked through the grades in order, youngest first, in every year.

Because a school's share of a program does not change from year to year, and because grade sizes barely change either, that produced the same answer every time. School R's running claim first crossed a whole student at second grade, and it did so in every year of the forecast:

Whole students assigned to School R — grades worked through youngest first, every year

Grade	2027-2028	2028-2029	2029-2030	2030-2031	2031-2032	2032-2033	2033-2034	2034-2035	2035-2036
0	0	0	0	0	0	0	0	0	0
1	0	0	0	0	0	0	0	0	0
2	**2**	**2**	**2**	**2**	**2**	**2**	**2**	**2**	**2**
3	0	0	0	0	0	0	0	0	0
4	0	0	0	0	0	0	0	0	0
5	0	0	0	0	0	0	**1**	**1**	**1**

The stripe down second grade is the visible problem. Read casually, it says something specific and actionable about second grade at School R. It says nothing of the kind. It is 21 rounding decisions all landing in the same row for the same arithmetic reason.

The tool begins each year's band at a grade chosen from the published tie-break setting and works around the band from there, so the starting grade varies from year to year and from program to program, but never between runs. The same students, the same totals, distributed differently:

Whole students assigned to School R — starting grade varied by year

Grade	2027-2028	2028-2029	2029-2030	2030-2031	2031-2032	2032-2033	2033-2034	2034-2035	2035-2036
KG	0	0	**1**	0	0	**1**	0	0	0

Grade	2027-2028	2028-2029	2029-2030	2030-2031	2031-2032	2032-2033	2033-2034	2034-2035	2035-2036
1	**1**	**1**	0	**1**	0	0	**1**	**1**	**2**
2	0	0	0	0	**1**	0	0	**1**	0
3	0	0	**1**	0	**1**	0	0	0	0
4	**1**	0	0	0	0	0	**2**	0	**1**
5	0	**1**	0	**1**	0	**1**	0	**1**	0

Every year's total is identical between the two tables, and every grade in every year still receives exactly the students who left it. Nothing about where students go has changed. The only difference is which box absorbs the rounding.

This is a presentation choice, not an accuracy improvement. Neither table is closer to the truth than the other; both are wrong by about half a student in every box, because the truth is half a student in every box. What the second avoids is a false pattern. A reader of the first table would reasonably conclude that second grade at School R is affected and the other grades are not. A reader of the second sees what is actually the case: a couple of students a year, somewhere in the elementary grades, with no particular grade implied.

8. The tool does show cohort patterns — when there is one to show

It is fair to ask whether a model that produces a flat result is capable of showing a cohort pattern at all. It is, and the contrast is easy to demonstrate.

Below are two runs of the same closure at the same school, moving 100 percent of grades K–5, with one difference. The first is marked immediate; the second is phased, meaning students already enrolled above the starting grade are grandfathered and finish where they are (Part 2, Section 3.4). Figures are simplified to a flat 30 students per grade so the pattern is unmistakable.

Immediate closure — students leaving, by grade and year

Grade	2027-2028	2028-2029	2029-2030	2030-2031	2031-2032	2032-2033
KG	**30**	**30**	**30**	**30**	**30**	**30**
1	**30**	**30**	**30**	**30**	**30**	**30**
2	**30**	**30**	**30**	**30**	**30**	**30**
3	**30**	**30**	**30**	**30**	**30**	**30**
4	**30**	**30**	**30**	**30**	**30**	**30**

Grade	2027-2028	2028-2029	2029-2030	2030-2031	2031-2032	2032-2033
5	**30**	**30**	**30**	**30**	**30**	**30**

Phased closure — same school, same 100 percent, same years

Grade	2027-2028	2028-2029	2029-2030	2030-2031	2031-2032	2032-2033
KG	**30**	**30**	**30**	**30**	**30**	**30**
1	0	**30**	**30**	**30**	**30**	**30**
2	0	0	**30**	**30**	**30**	**30**
3	0	0	0	**30**	**30**	**30**
4	0	0	0	0	**30**	**30**
5	0	0	0	0	0	**30**

The staircase in the second table is a cohort pattern, and the tool produces it without being asked, because under a phased change there genuinely is one protected group aging through while everyone above them is grandfathered.

Under an immediate closure there is no such group. Every grade departs in every year, so a flat result is the correct one. The stripe described in Section 7 was never a missing cohort pattern; it was rounding, and it has been addressed as rounding.

9. How to read the output

Three practical points follow from all of this.

- **Trust the school and year totals.** A receiving school's total arrivals in a given year are accurate. Across the current scenario runs, a receiving boundary's total from one closing program across the grade band lands within an average of 0.3 students of its exact share and never more than a student away; a boundary that receives no one had an exact share of less than one student.
- **Do not read grade-level redistribution figures as a grade-level forecast.** Which grade a returning student is assigned to is decided by whole-number arithmetic, not by evidence. This applies only to students arriving through a boundary-based redistribution; every other figure in the model, including PRC's baseline and all direct school-to-school transfers, is grade-specific in the ordinary sense.
- **Expect the grade to vary between years, and do not read meaning into the variation either.** A school that gains two students may see them in different grades in successive

years. That variation is not a projection of movement between grades. It is the visible form of a choice the data cannot make.

For anyone wanting to check a particular figure, the detailed output records the reasoning behind every number — the rule that produced it, the historical share applied, the exact fractional result, and whether it was rounded up or down. Part 2, Section 6 explains how to read it.

10. Why not simply use grade-level residency data?

The natural response to all of this is to ask for better data: if the residency shares were calculated for each grade separately rather than for a grade band, the model would know which grades the students belong in.

It would not, and the attempt would make matters worse. A program of roughly 25 students per grade draws from more than 40 boundary areas. Computing shares grade by grade means dividing 25 students among 40 areas, which produces counts that are almost entirely zeros and ones. Averaging three years of such figures does not produce a stable rate; it produces noise with the appearance of precision, and the model would then distribute students according to which boundary happened to have a child in a particular grade in a particular past year.

The grade band is the level at which this data is statistically meaningful. The honest course is to use it at that level and to be clear, as this part is, about what it can and cannot support.

11. Summary

When a program closes and its students return to their neighborhood schools, the model applies a historical residency rate to a forecast that has already accounted for cohort progression. It holds counts rather than students, so nothing within it ages from one grade to the next; the aging has already been done by PRC and is carried in the figures the model starts from.

For a school whose boundary supplied only a small share of a closing program, the resulting expectation is a fraction of a student in each grade in each year. Reporting that as whole children requires choosing grades for which there is no evidence. The tool varies that choice across the forecast rather than concentrating it in one grade, so that an arbitrary decision looks arbitrary instead of looking like a finding.

Totals by school and year remain accurate and can be relied on. Grade-level detail arising from a boundary redistribution should be read as an allocation, not a forecast.

Part 4. Boundary Modeling and Spatial Analysis

How proposed attendance boundaries become the counts in the scenario rules

A companion to Part 2 · October 2026

About this part: Part 2 explains how PRC's forecast is combined with a scenario's rules. Several of those rules carry a percentage for resident students: an attendance area split 55 percent to one school and 45 percent to another, say. This part explains where those percentages come from. It describes how proposed boundaries are drawn and stored, how the students who live in the affected blocks are counted, how the counts reach the rule table, and where the same block assignments are used again for the demographic and walking-distance measures. It is written for readers who want to verify that the work is residence-based rather than be told that it is.

A note on names and figures: As in Part 2, the examples use generic names (School A, Area 1) and simplified figures. The database objects are named as they exist, so that a reader with access to the GIS database can follow the path from a block to a number.

1. The question this component answers

Two of the three kinds of values in a scenario come from enrollment history. PRC's forecast, and the residency table, are used when a closed school or program returns students to their home areas. The third type of value comes from the map. When a scenario changes an attendance boundary, the share of a school's resident students who move with it is not a planning judgment. It is a count of the students who live in the blocks being reassigned, under the current boundary and under the boundary being proposed, divided by the students who live in the area today. Part 4 follows that count from the map to the rule table.

2. The data

2.1 Blocks

The unit of all boundary work is the 2010 census block. Every block in the district carries its census identifier (`GEOID10`), the planning area it falls in, and a school program assignment for each grade from kindergarten through 12th grade. A school's boundary in this system is not a drawn line; it is the set of blocks assigned to a school for a grade. Drawing a proposed boundary means reassigning blocks.

The block geography is old. The district's boundaries were drawn against these blocks when they were first created, and one boundary change since then, between Lincoln and Ida B. Wells-Barnett in 2016, cut across a block. The work-around for that divided block is described under residency below. PPS states the age of the geography plainly: While the block geography is dated, it is sufficient for modeling by school and year which is the same way scenarios are being compared. The block is the unit in which boundaries are drawn and counted. It does not describe individual homes or residential complexes.

2.2 Students

Each year's roster (`STU_Profiles_2913`) holds one row per student per school year, going back to 2007-2008, and includes the geocoded home location (state plane, SRID 2913), the census block it falls in (`Block_id`), the school and program the student attends, and the school boundary assignment in which the student resides at each grade band. This is the chain that produces it:

- Addresses come from the student information system, are processed into the enrollment profile table, and are geocoded by a scheduled PPS job. This job runs daily, and updates a geocoded-address table. An address the geocoder cannot place is carried as Undetermined until it can be geocoded.
- Residency is derived from the geocoded address point, not from the school of enrollment. A point inside the district's boundary polygons receives a resident school for each level; a point outside them is Out of District; a student with no point is Undetermined. The resident high school is taken from the school's boundary polygon rather than from the census block, because one high school boundary divides a block.
- A load procedure joins each point to its census block and rebuilds the spatial roster for every year. The roster also carries a second set of resident-school columns, ending in "`_living_2027`", updated in a one-time update from two GIS polygon layers: the K–8 boundaries adopted for 2023-2024, and the high school boundaries, which have not changed since 2021-2022. The one change already adopted for 2027-2028 is the end of a dual assignment at Jefferson High School. Those are the boundaries every scenario is drawn against and the residency table in Part 2 is built on them as well.
- `PRC_group`, the enrollment category PRC forecasts (including: residents attending the neighborhood program, non-residents attending it, each immersion program, special programs), is a computed column in the roster, derived from the resident school and the school of enrollment. The groups the forecast is organized by, and the groups the GIS counts, are therefore one definition, not two.

2.3 School programs

`sdss_SCHOOL_PROGRAMS` lists every program that can be assigned to a block: a program id, the school program name (the same names the forecast uses), the facility code that links it to the school site, and the neighborhood id used to select a whole attendance area at once.

2.4 Assignments, scenarios and settings

Four tables carry the boundary work:

- `SDSS_Block_Grade_Assignments` is the working layer. This table has one row per block and one column per grade (`gr00` to `gr12`) holding the program id assigned. This is the layer edited in ArcGIS Pro. A spatial copy (`SDSS_Block_Grade_Assignments_2913`) is kept up to date by a programmatic trigger and is what the maps and the spatial joins read.
- `SDSS_Scenario_Block_GRADE_Assignments` is the archive and contains one row per block, grade, and scenario, along with the date of the last change and the user who made the change.

- `SDSS_scenarios` is the registry and includes scenario number, scenario name, and a locked flag.
- `SDSS_Parameters` is a single-row settings table (a trigger refuses a second row or an empty table) which contains the active scenario, the comparison scenario, the school and grades currently being assigned, the list of selected blocks, and the enrollment year the counts use.

3. Drawing a proposal

The Enrollment Balancing Toolbox (`EnrollBal.pyt`) runs inside ArcGIS Pro and reaches the GIS database through a small helper (`db_helper.py`) that calls the stored procedures named below. Its tools count the students in a map selection, assign the selected blocks to a school for chosen grades, manage scenarios, preview a scenario, and generate a report. Everything the toolbox does is a database call, so every operation can be repeated and audited in SQL independent of ArcGIS Pro.

Scenarios. A scenario is created with Save As (`SDSS_SProc_Add_Scenario`), which registers a new name and number and copies the block-by-grade assignments of an existing scenario into the archive. Choosing a scenario (`SDSS_SProc_Choose_Scenario`) writes its name and number into the settings row, rebuilds the comparison table described in Section 4 and refreshes the school site points; a trigger on the settings row (`sdss_TRIG_Scenario_Parameter`) then copies the scenario's archived assignments into the working layer, pausing the archive trigger so that the load is not written back (Figure 2a). The map the analyst looks at is this working layer based on the scenarios assignments and nothing else. The census blocks in the map are symbolized by the program id in one grade column, so changing the active scenario in the settings row replaces the program ids in the grade columns and redraws the map. No boundary is drawn or stored anywhere as a line; what looks like a boundary on the screen is the edge between blocks that carry different program ids. A locked scenario cannot be edited. The triggers that write to the archive check the flag and do nothing for a locked scenario, which is how a published scenario is protected while the next one is drawn.

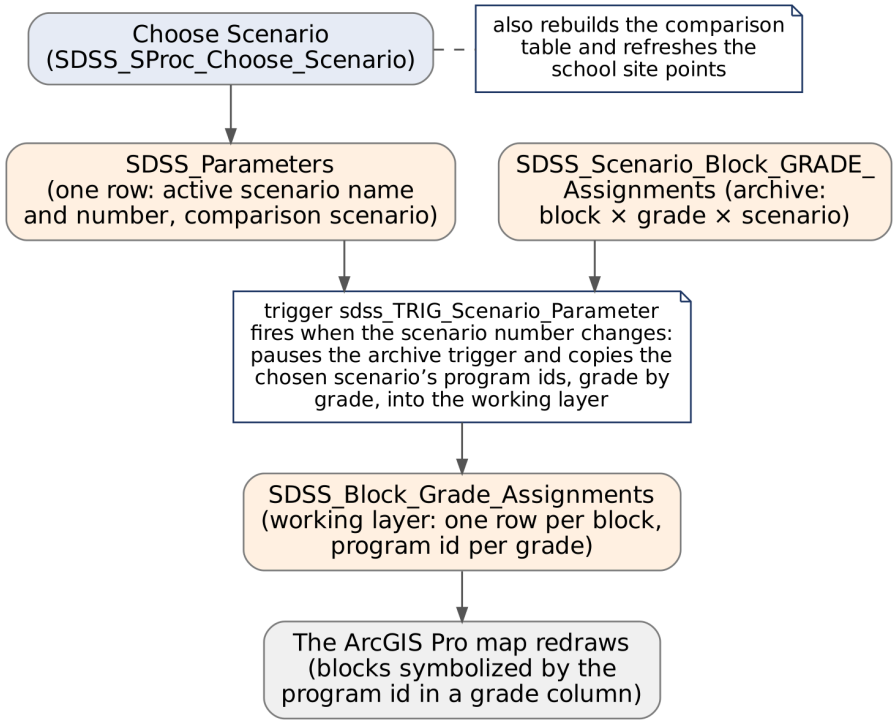


Figure 2a. Choosing a scenario. The Choose Scenario tool writes the scenario's name and number into the settings row. A trigger on that row copies the scenario's archived program ids into the working layer, grade by grade, with the archive trigger paused so that the load is not written back. The map is the working layer, so it redraws.

The comparison scenario. The official boundaries are themselves a scenario, called the comparison scenario. Once a year a stored procedure (SDSS_Parameters_update_Scenario_Name_Compare) names the official boundaries as the comparison scenario and writes onto every geocoded address the neighborhood school, by grade, that the scenario's blocks assign. Each student then has two records of their resident school: this block-based one on the address, and the polygon-based one on the roster (Section 2.2). The two are compared each year when the roster is loaded. They should differ only at the adopted high school change and for students with no in-district address point. Because the comparison scenario is the official boundary, and the 2027-2028 columns are that boundary plus the one adopted change, the residency groups in the forecast and the block counts describe the same areas.

Assigning blocks. The analyst selects blocks on the map and chooses the school and the grades. The selection is passed to the database as a list of block ids; the procedure (`SDSS_SProc_update_block_grade_assign_schools_by_csv_list`) writes the chosen program into those blocks for the chosen grades in the working layer. Two triggers fire on that update: `SDSS_TRIG_Scenario_update_from_block_grade_assignments` copies each changed grade into the archive for the active scenario, stamped with the date and user, and `SDSS_TRIG_Sync_Block_Grade_Assignments_2913` refreshes the spatial copy, from which the map layers in Figure 2b are drawn. A whole attendance area can be moved in one step: a procedure selects every block whose kindergarten assignment is a given school and reassigns the K–5 grades (a counterpart does

the same for 6–8), which is how a closing school's entire area is handed to a successor before a split is drawn.

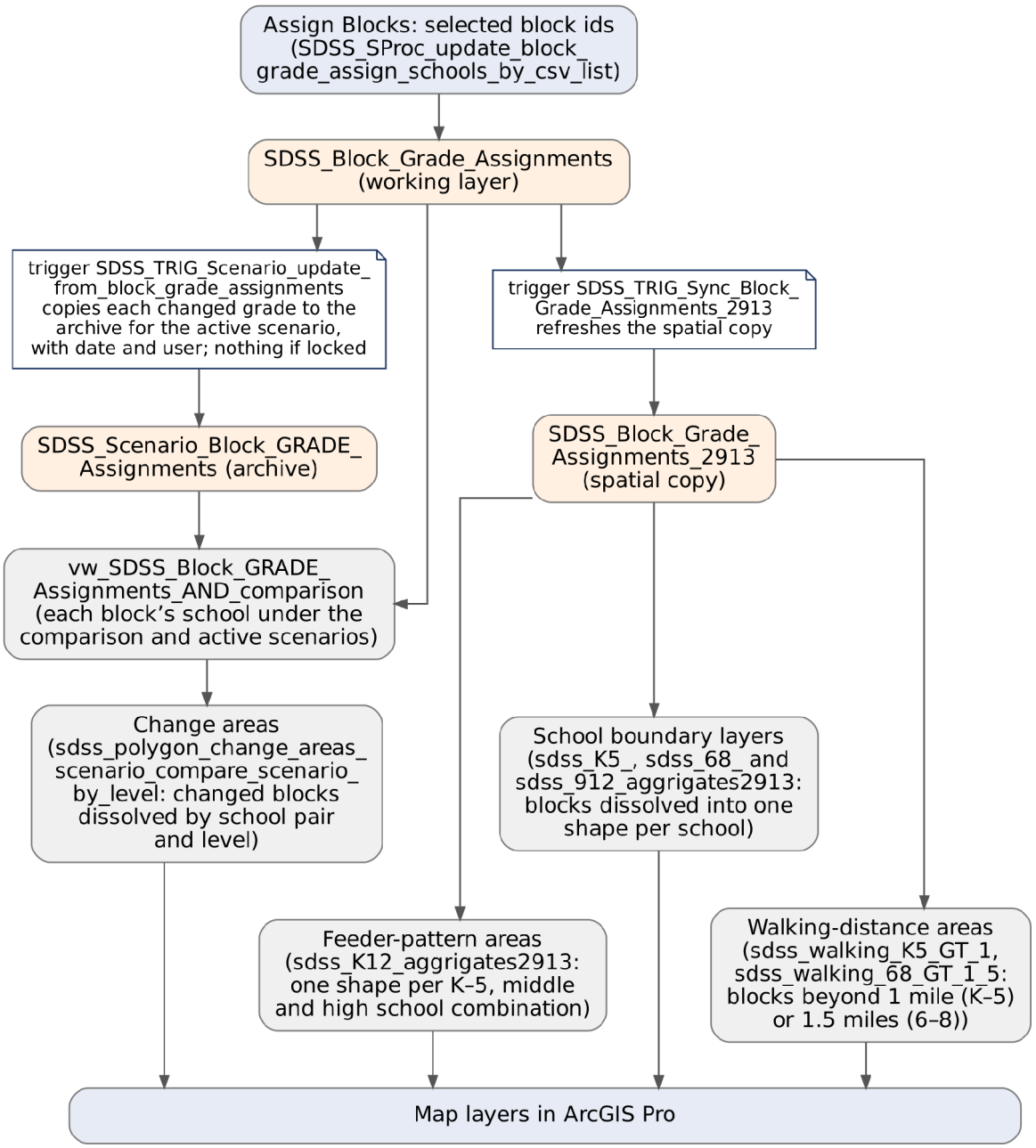


Figure 2b. Assigning blocks, and the map layers built from the assignments. Assigning blocks writes new program ids into the working layer; one trigger copies each change into the archive for the active scenario (unless it is locked) and another refreshes the spatial copy. Views on these tables are the layers drawn on the maps: the change areas, from the block-by-block comparison with the comparison scenario, and the school boundary, feeder-pattern and walking-distance layers, each made by dissolving blocks into larger shapes.

Because assignments are per grade, a block can send its K–5 residents to one school and its 6–8 residents to another, or a block within a K–8 school can keep its K–5 area while its middle grades are reassigned. The same mechanism carries the first-year form of a proposal: a boundary scenario can be stored twice, once as the end state and once as the first year only, so that first-year demographics can be computed on first-year boundaries.

4. Counting

4.1 Change areas

A comparison view (`vw_SDSS_Block_GRADE_Assignments_AND_comparison`, with a twin on the spatial copy, `vw_SDSS_Block_GRADE_Assignments_AND_comparison_2913`) lists, for every block, its school under the comparison scenario and under the active scenario, grade by grade; a procedure writes it to a table for speed whenever the scenario changes. A second view (`sdss_polygon_change_areas_scenario_compare_scenario_by_level`) dissolves the blocks whose assignment differs into one polygon per pair of schools and per level (K–5, 6–8, 9–12). That is the change-area layer drawn on the maps (Figure 2b).

4.2 Residents in a selection

For any list of blocks, a table-valued function (`SDSS_Count_comparison_grade_group_csv`) returns by grade group and by the pair of comparison school and scenario school: the number of students living in those blocks, the number of them attending their neighborhood program, and each as a proportion of the comparison school's total.

4.3 Residents by pair, across eleven years

The summary view (`vw_SDSS_Assignments_AND_comparison_summary_2913`) does the same for every pair of comparison school and scenario school, with no selection, and for every school year from 2015-2016 through 2025-2026: the count of residents attending the neighborhood program who live in the blocks the pair covers, and the share of the comparison school's residents they represent in that year. A three-year aggregate (2022-2023 through 2024-2025) is carried alongside. The eleven-year series is the stability check. A split that reads 55 percent today and ranged between 52 and 57 percent over the decade is a different kind of number from one that swung between 35 and 70, and the series shows which kind a proposal rests on.

4.4 From the count to the rule

The percentage on a resident rule in the scenario table is this share: the fraction of the comparison school's residents attending the neighborhood program who live in the blocks the scenario sends to the receiving school, read from the current year and checked against the series. The percentages are the output of an iteration between a planner and an analyst. The planner proposes a split, the analyst draws it and counts, and either the line or the intent is revised until the count serves the area's goals. They are not starting assumptions.

Worked example (illustrative only). School A closes and its K–5 area is divided: the blocks north of a main street go to School B, the rest to School C. The working scenario assigns them. The count returns, for grades K–5 in the current year, 120 residents attending the neighborhood program at School A, of

whom 66 live in blocks now assigned to School B and 54 in blocks assigned to School C. The summary view shows the north-side share at 55 percent this year and between 52 and 57 percent in each of the previous ten. The rule table records School A's residents, grades K–5, as 55 percent to School B and 45 percent to School C, and Part 2 takes it from there: PRC's forecast of School A's residents is split in those shares in every forecast year, and School A's boundary is carried forward to Schools B and C in the residency table in the same shares (Part 2, Section 3.3).

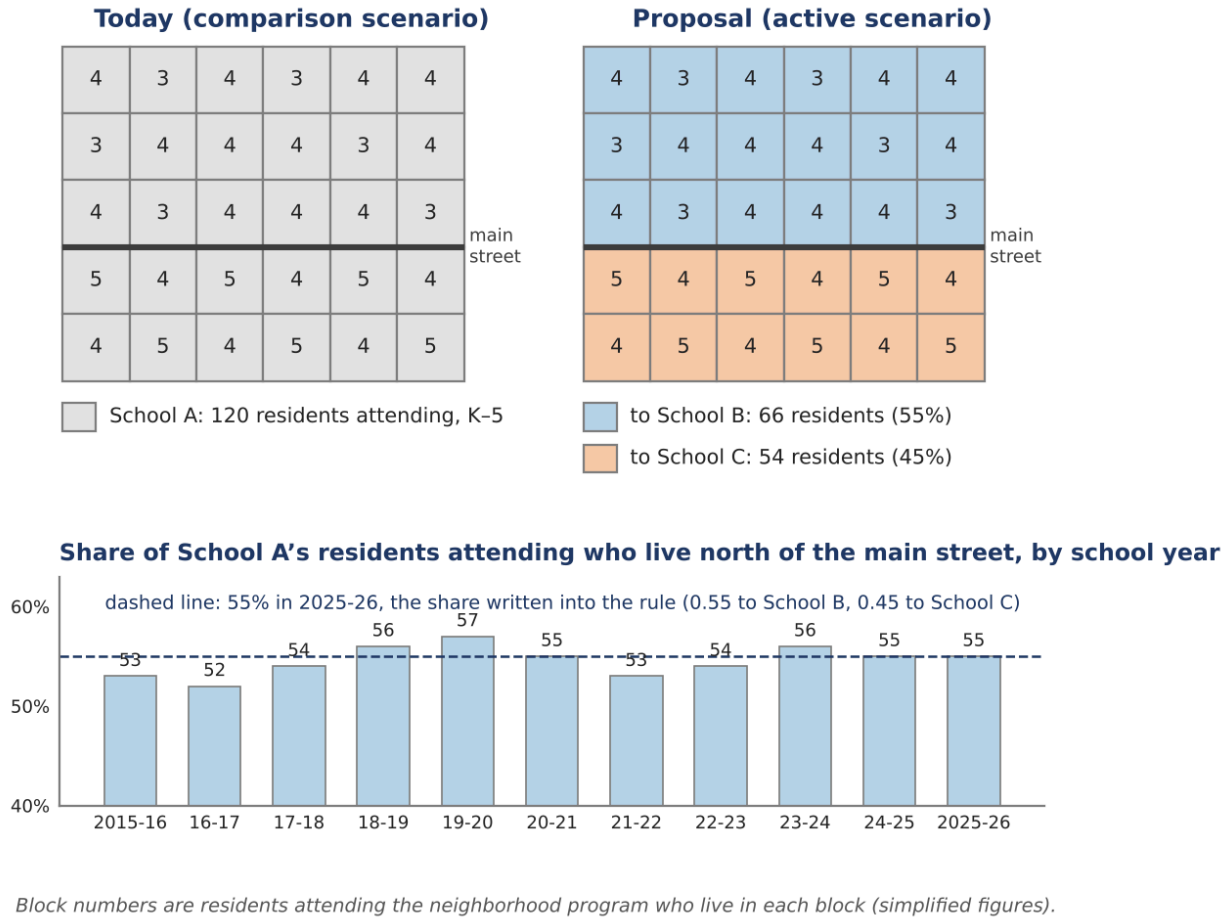


Figure 3. The worked example. Top: the blocks of School A's K–5 area under the comparison scenario and under the proposal, with the residents attending who live in each block. Bottom: the north-side share in each school year as the summary view reports it; the current year's 55 percent is the share written into the rule.

5. Where the GIS work enters Scenarios A and B

The modeling tool described in Part 2 does no spatial work of its own. It consumes two tables built here, and the student-level measures read the block assignments directly. The three hand-offs:

What the GIS work produces	Where it lands	How it is applied
The share of a school's residents attending the neighborhood program who live in the blocks a proposal reassigns (Section 4)	The percentage on each resident rule in the scenario rule table	A share rule: that fraction of PRC's forecast of the school's residents is moved to the receiving school in every forecast year (Part 2, Section 3.5); at a closure, the school's boundary is carried into the residency table in the same shares (Section 3.3)
The residency table: where each school's students live, by 2027-2028 attendance area (Section 5.2)	The Proportions tab of the scenario workbook	Every rule whose destination is "neighborhood": the departing students are split across the areas in the table's shares, after the table is corrected for schools the scenario closes (Part 2, Sections 2.3, 3.3 and 3.6)
The block-by-grade assignments of each scenario	The scenario archive, read by the student-level procedure	Each enrolled resident is placed at the school assigned to their block and grade; the demographic and walking measures follow (Sections 5.3 and 5.4)

5.1 The resident rules

In the published rule table, Scenario A carries 43 rules that move residents attending the neighborhood program to a named school, 25 of them at a share below 100 percent; Scenario B carries 39, 23 of them below 100 percent. Every one of those shares is a block count from Section 4: a 100 percent rule is an attendance area handed whole to a successor, and a share below 100 percent is a split, read from the comparison count and checked against the eleven-year series. The rule table is published with the scenarios, so each share can be traced to the blocks it came from with the objects named in this part.

5.2 The residency table

The second table is built from the roster by a query rather than from the map. For every school and program of enrollment, every PRC group and every grade band (K–5, 6–8, 9–12), it gives the share of that program's students who live in each attendance area, where the area is the student's `es_living_2027`, `ms_living_2027` or `hs_living_2027` school for the band, and Out of District where there is none. The query computes the share separately for each school year in its window, the years since the K–8 boundary change of 2023-2024 (2023-2024 through 2025-2026 at the time of writing), averages the yearly shares with equal weight, and renormalizes so that each program's shares add to 100 percent across its areas. Pre-kindergarten is excluded. The result is loaded into the Proportions tab of the scenario workbook, 2,669 rows covering 101 schools and programs and 76 areas.

The modeling tool reads that tab for every rule whose destination is “neighborhood” rather than a named school: 22 rules in Scenario A (14 for transfer students at closing schools, 6 for immersion programs that move, 2 for special programs) and 20 in Scenario B. Before each scenario runs, the tool prepares its own corrected copy of the table in which every area belonging to a school the scenario closes is carried forward to that school’s successors in the shares of the resident rules (Part 2, Section 3.3), so the two GIS products are consistent with each other: a closed area is split the same way in the residency table as on the map. Because the areas in the table come from the 2027-2028 living columns and the comparison scenario is the same boundary set, the residency table and the block assignments describe the same areas.

5.3 The demographic pages

The student-level procedure (`sdss_student_data_w_scenario_assignments_INSERT`) is described in Part 5 and shows for status quo and each scenario each student’s assignments. For every student with a usable point it reads the block assignment under each scenario from the same archive table, at the student’s own grade, and that is the school a resident is shown at when a boundary changes. The procedure requires that every scenario in the rule table have block assignments under exactly the same name in the GIS registry; a scenario with none is reported as an error, not returned as a quiet result. Part 5 describes the placement rules in full. Student data in the table can be aggregated then by demographics to describe the compositions of schools under a scenario.

5.4 Walking distance

A table (`SDSS_Walking_Distance_Time`) holds the walking distance in feet and the time in minutes from each census block to each school site: an origin–destination list generated in ArcGIS around 2016. The one-mile threshold for K–5 and the one-and-a-half-mile threshold for 6–8 are the district’s transportation thresholds, and two views (`sdss_walking_K5_GT_1` and `sdss_walking_68_GT_1_5`) dissolve the blocks beyond each threshold into one polygon per school for the map (Figure 2b).

The published measure comes from a second view (`sdss_student_data_w_scenario_assignments_chg_walking`). It joins the student-level scenario assignments of Section 5.3 to the distance table twice: once for the distance from the student’s block to the school attended today, and once for the distance from the same block to the school the scenario assigns. It covers students who attend their neighborhood program and have a block. For each receiving school under each scenario, and for the district as a whole, it reports the number and the share of those students more than a mile and more than a mile and a half from the assigned school. It also shows the median, mean, and quartiles of the change in walking distance. A Status Quo row carries the same measures against the school attended today, so every scenario is read against one baseline. Two consequences follow. The measure counts residents attending the neighborhood program only, so a scenario that moves only transfer or immersion students does not change it. And because it reads the student-level assignments, it carries that model’s whole-group placement rules rather than the enrollment model’s shares.

6. Limits

- **A block is the smallest unit.** A boundary that divides a block cannot be represented at the block level. In the one instance where a block is divided, the polygon layer governs the student's residency and the block layer governs the scenario, and the two are reconciled by hand.
- **Students without a usable point get no block and no scenario assignment.** They are carried as "Undetermined". They are few, and their number is checked at each load.
- **The two all-immersion elementary areas.** Rigler and Lent are only Spanish Immersion schools, and their attendance areas work as dual-assignment areas. A family living in either is guaranteed an immersion place or otherwise attends a neighboring school's English program, Scott's for the Rigler area and Marysville's for the Lent area. For residents and non-residents alike, PRC forecasts incoming students with its own method, and the forecast PPS receives is by school, enrollment group, grade, and year; it does not include a forecast of the students living in each attendance area. PRC counts the two areas as Scott's and Marysville's and forecasts Rigler and Lent as immersion only. The model reassigns each neighborhood school's residents by the scenario's boundaries and, in Scenarios A and B, moves immersion students to the immersion hubs. In the district's GIS, however, the two areas are still drawn as Rigler's and Lent's own attendance areas rather than as Scott's and Marysville's. That difference has two small effects on the published scenarios. The return-home step reads the residency table built on those layers, so a few returning students who live in the two areas are placed in a neighborhood program at Rigler or Lent that PRC does not forecast: 5 in 2027-2028 in each of Scenarios A and B, and no more than 13 in any year. And the student-level demographic and walking figures at Scott and Marysville treat those schools' students from the two areas as living in Rigler's or Lent's area. A baseline that maps the two areas to Scott and Marysville has been built for the next iteration.
- **Planned housing is not in the blocks.** Block counts and the residency table describe where students live today. PRC's forecast carries development at the school level, so a scenario splits a housing-aware total by housing-blind shares. The question is within the scope of PRC's review.
- **The eleven-year series crosses boundary changes.** The K–8 boundaries changed for 2023-2024, so the early years are read as a check on stability, not as current shares. The view's three-year aggregate (2022-2023 through 2024-2025) is a different window from the residency table's (2023-2024 through 2025-2026).
- **The walking distances are from 2016.** The origin–destination list was generated once, in ArcGIS, around 2016, and has not been regenerated. Streets and crossings change slowly, and the list serves the same school-and-year purpose as the block geography; it should be read with the same caution.
- **Very small reassignments are real.** A handful of students per grade who live much closer to a neighboring school may be assigned there deliberately. Such a move shows in the rule table as a share of a few percent and is not an error.

7. Checks

- One program per block, grade, and scenario. The student-level procedure stops on a duplicate rather than counting a student twice.
- Every scenario in the rule table has assignments in the GIS registry under exactly the same name.
- Block counts reconcile to the roster. The residents attending the neighborhood program counted by block for a school equal the roster's count for that school and year.
- Every change to an assignment carries the date and the user who made it. A locked scenario cannot change.
- The settings row cannot be duplicated or emptied, so there is always exactly one active scenario and one comparison scenario.
- At the annual roster load, the polygon-based living school on the roster and the block-based neighborhood school stamped on the geocoded address are compared; they are expected to differ only at the adopted high school change and for students with no in-district point.

8. Object reference

The objects named in this part, in the GIS database, by role.

Object	Role
`STU_Profiles_2913`	Annual student roster with geocoded location, census block, school of enrollment, living school by level (current, and the 2027-2028 set the scenarios are drawn against) and the computed PRC group.
`REV_ADDRESS_GEOCODED`	Geocoded addresses, with the neighborhood school by grade stamped from the comparison scenario's block assignments.
Residency proportions query → Proportions tab	Share of each school and program's students living in each 2027-2028 attendance area, by PRC group and grade band, averaged over the school years since 2023-2024; read by the modeling tool for every "neighborhood" destination.
`sdss_SCHOOL_PROGRAMS`	Every assignable school program: id, name, facility code, neighborhood id.
`SDSS_Block_Grade_Assignments`	Working layer edited in ArcGIS Pro: one row per block, one program id per grade.
`SDSS_Block_Grade_Assignments_2913`	Spatial copy of the working layer, kept in step by trigger.
`SDSS_Scenario_Block_GRADE_Assignments`	Archive: one row per block, grade and scenario, with date and user of the last change.

Object	Role
SDSS_scenarios	Scenario registry: number, name, locked flag.
SDSS_Parameters	Single-row settings: active scenario, comparison scenario, school and grades being assigned, selected blocks, enrollment year.
SDSS_SProc_Add_Scenario	Save As: registers a scenario and copies another scenario's assignments into the archive.
SDSS_SProc_Choose_Scenario	Makes a scenario active: writes it into the settings row, rebuilds the comparison table and refreshes the school site points.
SDSS_SProc_update_block_grade_assign_schools_by_csv_list	Assigns the selected blocks to the chosen program for the chosen grades.
SDSS_TRIG_Scenario_update_from_block_grade_assignments	Copies each changed grade of a block into the archive for the active scenario, unless it is locked.
SDSS_TRIG_Sync_Block_Grade_Assignments_2913	Refreshes the spatial copy of the working layer.
sdss_TRIG_Scenario_Parameter	On a change of active scenario, copies that scenario's archived assignments into the working layer, with the archive trigger paused.
SDSS_Parameters_update_Scenario_Name_Compare	Names the official boundaries as the comparison scenario and stamps geocoded addresses with their neighborhood school by grade.
vw_SDSS_Block_GRADE_Assignments_AND_comparison	Per block and grade: school under the comparison scenario and under the active scenario; written to a table when the scenario changes, and read by the change-area view.
vw_SDSS_Block_GRADE_Assignments_AND_comparison_2913	The same comparison on the spatial copy.
sdss_polygon_change_areas_scenario_compare_scenario_by_level	Change areas dissolved to polygons by pair of schools and level.
sdss_K5_aggrigates2913, sdss_68_aggrigates2913, sdss_912_aggrigates2913	School boundary layers: the spatial copy's blocks dissolved into one shape per school, read from grades 2, 7 and 10.
sdss_K12_aggrigates2913	Feeder-pattern areas: one shape per combination of K–5, middle and high school.
sdss_walking_K5_GT_1, sdss_walking_68_GT_1_5	Blocks more than a mile (K–5) or a mile and a half (6–8) from their assigned school, dissolved by school for the map.

Object	Role
SDSS_Count_comparison_grade_group_csv	Residents in a selection of blocks by grade group and school pair, with proportions of the comparison school.
vw_SDSS_Assignments_AND_comparison_summary_2913	Residents attending the neighborhood program by school pair for each school year 2015-2016 through 2025-2026, with proportions.
sdss_student_data_w_scenario_assignments_INSERT	Places each enrolled student under each scenario using the archived block assignments.
SDSS_Walking_Distance_Time	Origin–destination list of walking distance and time from each block to each school site, generated in ArcGIS around 2016.
sdss_student_data_w_scenario_assignments_chg_walking	For residents attending the neighborhood program: students more than a mile and more than a mile and a half from the assigned school, and the median, mean and quartiles of the change in walking distance, by receiving school and scenario and for the district, with a Status Quo baseline.

Part 5. Demographic Impact Model

Student-level scenario assignment and next-year roll-forward

A companion to Part 2 · October 2026 · Procedure:
`GIS.dbo.sdss_student_data_w_scenario_assignments_INSERT`

1. Purpose

Rightsizing scenarios are evaluated in two complementary ways. The PRC enrollment model forecasts how many students would attend each school under each scenario. The model described here answers a different question: which of today's students would be affected, and how. It takes every currently enrolled student, determines where that student would be assigned under each scenario, and compares the result with where they would be under the status quo. Because it works with real students rather than projected counts, its results can be summarized by demographic group and by geography.

This model reports composition, not enrollment. Its counts are today's students redistributed under a scenario. They are not a forecast, they do not include next year's kindergarteners or any in-migration, and they will not match the PRC enrollment model. The two models are built for different questions and should be read side by side, not reconciled.

The model produces two location assignments for every student and every scenario:

- **This year's assignment.** Where the student would be assigned at their current grade if the scenario were fully implemented. This is the original demographic-impact output.
- **Next year's assignment (the roll-forward).** Where the student would be one grade older, under the status quo and under the scenario. This captures the transitions that drive most of the real movement: 5th graders entering middle school, 8th graders entering high school, and immersion strands moving between buildings.

2. Inputs

Four sources feed the model. Two of them are primary for different kinds of change, and the distinction matters throughout.

Source	What it provides	Primary for
Scenario workbook (Google Sheet, loaded to SQL Server)	One row per scenario, school, PRC group, and grade band, giving the share of that group moving to a destination school and group. A destination of "neighborhood" means the program dissolves and students revert to the school their address is zoned to.	Program changes: immersion strands, special programs, closures of a program rather than a building.

Source	What it provides	Primary for
GIS boundary scenarios (ArcGIS Pro, Enrollment Balancing Toolbox)	For each scenario, the school each census block is zoned to at each grade. Drawn block by block, because demographics vary block to block and one boundary is usually split several ways.	Neighborhood boundary changes for residents.
Closed School Successors tab	The primary receiving school when a building closes, by grade band.	Transfer students at a closing school who have no boundary to fall back on.
Student roster (STU_Profiles_2913)	Every enrolled student for the modeled school year: campus, PRC group, grade, census block, living schools under the 2027 boundary set at each level, and demographics.	Who the students are and where they live.

Students are classified by **PRC group**, the same grouping the enrollment model uses: Residents Attending Neighborhood Program, Non-residents Attending Neighborhood Program, each immersion strand (Spanish, Japanese, Mandarin, Russian, Vietnamese), and Special Programs (focus options, charters, alternative programs, and similar). The group decides which rules apply to a student.

All "living" schools come from the **2027 boundary set** (the columns suffixed _2027). Boundary changes already scheduled for next year are built into that set, and every scenario is drawn against it. The un-suffixed living columns reflect boundaries that are about to change and are not used.

3. Population and scope

- **This year:** all students enrolled in grades K-12 in the modeled school year (pre-kindergarten excluded). Each student appears once per scenario.
- **Next year:** continuing students only, meaning this year's grades K-11. Next year's kindergarteners are not on the roster, so their demographics cannot be modeled. This year's 12th graders are expected to graduate, and it can’t be projected which ones do not graduate.
- **School years:** the procedure runs for the latest roster year by default and can run for a prior year in the same call, publishing each year separately. Scenario definitions and boundaries are not year-specific; only the roster changes.

4. Modeling principles

A handful of rules govern every decision in the model. They are stated once here because the same choices recur in both potential outcomes.

A student cannot be split

The enrollment model can send 60% of a program to one school and 40% to another. A student cannot be divided, so where a workbook row moves part of a group, the whole group follows the plurality destination. Ties are broken deterministically (by share, then by name) so repeated runs on the same

inputs produce identical results. A partial reassignment, below the 99% threshold that defines a full move, leaves students in place.

Two sources of truth

Residents of a neighborhood program follow their census block in the GIS layer, whatever the modeling tool workbook says about that school; the workbook's share for residents is informational. Students in immersion strands and special programs follow the modeling tool workbook, because the boundary layer says nothing about programs. This split is what lets a scenario redraw a boundary and move a strand independently.

First match wins, and every placement is explained

Rules run in a fixed order and each fills only students not yet placed. Every rule writes a note naming itself, so any count in a dashboard can be traced back to the rule and the inputs that produced it. The distribution of students across rules is recorded on every run and compared with the previous run, which is the primary signal that a change to the workbook or the procedure has moved something.

The status quo and the scenario are built the same way

For next year, the status-quo columns and the scenario columns are produced by the same rules from the same inputs, differing only in which boundary set and which program locations they read from. The output is a matched pair, the status quo and the scenario, built to be compared. It is not a forecast of either. For transfer students that the rules cannot see (Benson, da Vinci, special education placements, hardship transfers, lottery outcomes), the model places them in the same place on both status quo and scenario and contributes nothing to the difference, which is the intended treatment of anything unmodeled. The one thing that would bias the comparison is a rule applied to the status quo or the scenario but not both so every next-year rule is written as a pair.

5. This year's assignment

For each scenario, every student is assigned a school and group at their current grade by a cascade of twenty-five rules. The rules fall into the categories below; the full list is in Appendix A.

Category	What it does	Reads
Immersion program moved	A workbook row moving an immersion strand to a named school takes the whole strand there. Runs first so the workbook's authority over immersion is preserved.	Workbook
Program dissolved to neighborhood	A program whose workbook destination is "neighborhood" closes, and its students revert to the school their block is zoned to under the scenario. Honored only when the program is fully reassigned and neighborhood is the plurality destination, the same test a named destination must pass.	Workbook, then GIS

Category	What it does	Reads
Boundary change for residents	A resident whose block is zoned to a different school under the scenario moves there. Residents of a fully reassigned school go to their scenario boundary school regardless of the workbook's named destinations.	GIS
Named full reassignment	A school and group fully reassigned by the workbook go to the largest named destination.	Workbook
Untouched by the scenario	Students at schools and in groups no row or boundary change affects stay where they are. Most students in most scenarios.	Workbook and GIS, by absence
Transfers and out-of-district students	Non-residents at a school that stays open keep their place. Out-of-district students, who have no boundary, follow the largest named destination of their school and group.	Workbook
High school students	Grades 9-12 continue at their current school, reflecting the district rule that secondary boundary changes apply to incoming students. The model therefore shows little high school movement even where a scenario redraws high school boundaries.	Policy
Successors	A transfer student whose assigned school itself closes is redirected to that school's primary successor; a student at a fully reassigned school with no in-district boundary and no named destination goes to the successor or is recorded as out of district.	Successors tab

School names that differ between the GIS layer and the student system (Jefferson, Bridger) are normalized once after the cascade so that lookups keyed on a school name behave the same on both sides.

6. Next year's assignment

6.1 One set of placement rules for both status quo and scenarios

Each student receives two next-year placements:

- **Status quo:** NY_Campus_Enrolled and NY_PRC_Group, derived from where the student is enrolled today under current boundaries and program locations.
- **Scenario:** NY_Assignment_School and NY_Assignment_Group, derived from this year's scenario assignment under the scenario's boundaries and program locations.

Every rule below has an "a" form that sets the status-quo columns from the enrolled campus and PRC group, and a "b" form that sets the scenario columns from the scenario school and group, joined to the scenario. Each side is first-match-wins on its own null test, so the two can diverge for the same student, which is the point. A note on each side records the rule that placed the student.

6.2 Grade progression

Next year's grade is this year's grade plus one for every student in grades K-11. It does not depend on which school the student attends, so it is set once before any placement rule runs.

6.3 Moving up in place

A student moves up a grade at the same school when next year's grade falls within the span of grades that school serves for the student's group. The gradespan is the minimum to maximum grade enrolled (status quo) or assigned under the scenario (scenario side), not "someone is enrolled one grade up", so a sparse group with an empty grade in the middle still carries forward. The student's own grade is irrelevant: a 5th grader at a K-8 stays, a 5th grader at a K-5 does not, and K-8 schools need no special treatment.

Three groups inherit a wider gradespan than their own enrollment would show:

- **Non-residents** inherit the residents' gradespan at their school. Under board policy a non-resident may stay through the school's top grade unless the school closes; their own gradespan is narrow only because there are few of them.
- **Special programs** inherit the school's full gradespan. A charter or focus option is its own campus, so nothing changes; a co-located program continues at its host through the host's top grade.
- **Immersion strands** inherit the school's full gradespan, so a program still growing into its final grade continues to 12, such as McDaniel's Vietnamese strand, grades 9-11 today . To keep a K-5 strand at a K-8 building from being carried to 6th grade in place, a student whose school, group, and grade have an immersion feeder row (6.4) is excluded from this rule and leaves instead.

On the scenario side the gradespan is read from scenario assignments. So for a K-8 where the scenario converts to K-5, the school has no one assigned above grade 5, and does not carry its 5th graders forward. In a scenario where the K-5 converts to K-8, the 5th grader does carry forward.

6.4 Immersion feeder patterns

Immersion strands move between buildings in fixed patterns that the boundary layer does not describe. A K-5 strand feeds one middle school and a middle school strand feeds one high school. These patterns are held in a single table (Appendix C) and applied in five steps, each reached only if the step before did not place the student:

1. **Status quo.** For a student whose enrolled campus, group, and grade match a row, they are assigned to that row's destination, in the same group.
2. **Scenario.** The same immersion feeder pattern lookup on the scenario school and group, with one additional condition: the destination must still have the strand one grade up in that

scenario. If the scenario moved or closed the program at the destination, the student is not assigned there.

3. **Program moved by the scenario.** If the destination's strand was fully moved by the workbook at next year's grade, the student follows it to the workbook's named destination, in the workbook's group.
4. **Strand relocated by the scenario.** If the scenario closed or moved the student's own school, so that the scenario school matches no feeder row, the student still follows the feeder pattern of the school attended today, provided they remain in the strand and the destination still has that strand one grade up. A strand dissolved into the neighborhood program is excluded and goes through the neighborhood rules instead.
5. **Strand at the student's own school moved at the next grade.** A K-8 strand whose 6-8 portion is relocated in the scenario is covered by none of the above. The school's scenario gradespan stops at grade 5, a K-8 has no feeder row, and any feeder destination that a merged-in strand brought with it may also have moved. If the workbook fully moves the strand at the student's own scenario school at next year's grade, the student follows it to the workbook's destination. This is step 3 applied to the school itself rather than to a feeder destination.

The feeder table is checked against the roster on every run source row that matches no students, or a destination with nobody in that group one grade up today, is reported. Several corrections to the table, and the fifth step above, came from those checks and from the comparison scripts in section 8.

6.5 If a Spanish Immersion program has no one-to-one feeder HS

At grade 8 a student goes to the Spanish Immersion program serving their high school attendance area, which a crosswalk maps from the HS boundary school to the HS that houses the program (Appendix C). The crosswalk was read off the roster as the most common high school attended by current 10th-grade Spanish Immersion students in each attendance area. The status-quo side keys on the student's HS living school under the 2027-2028 set; the scenario side keys on the HS boundary school of the student's block under the scenario, with the same "program still there" condition as the feeders.

6.6 Neighborhood transitions

Everyone at a transition grade who is not in an immersion strand goes to the school serving their address at the next level, as a resident: 5th graders to the middle school, 8th graders to the high school. The status-quo model reads the roster's living school at that level under the 2027-2028 set; the scenario model reads the block's school at that level under the scenario, taken from the GIS layer at a fixed anchor grade (grade 7 for middle school, grade 10 for high school) rather than the student's own grade.

This rule includes non-residents because a transfer does not carry to the next level, as well as special-program students whose program ends at the transition grade. A student with no living school (out-of-district or undetermined address) has nowhere to fall back to and is left unplaced on that model.

6.7 The change flag

For every K-11 student, NY_change is 1 when the scenario's next-year school differs from the status quo's, and 0 when they are the same. A student whose next-year school is unknown under both the status

quo and the scenario is not a move and is marked 0. A student placed on one side and not the other is 1, so that anything the rules could not resolve stays visible as a difference rather than disappearing into the unchanged. The flag compares schools only; a student who stays in the same building in a different group is 0.

7. What the model does not do

- **Forecast.** It does not show cohort-survival ratios, in-migration, attrition, or new kindergarteners. Every count is a subset of today's students.
- **Split students.** A 50/50 workbook split is modeled as 100% to the plurality destination. For residents this rarely matters because they follow the GIS block; for programs it means a split strand is shown entirely at one school.
- **Predict individual transfers.** Lottery programs, hardship transfers, special education placements and similar are not predicted. They are treated identically on both status quo and scenario and so do not affect the comparison.
- **Move current high school students.** Grades 9-12 are held at their current school by policy, so scenarios that redraw high school boundaries show movement only through next year's 9th graders.
- **Place students with no address.** Out-of-district and undetermined-address students at a transition grade have no living school to fall back to and are left unplaced on both sides, which the change flag treats as no change.
- **Place a dissolved strand next year.** Where a scenario dissolves an immersion strand into neighborhood schools, its 5th and 8th graders go through the neighborhood rules as residents rather than following a program.

8. Validation and quality assurance

8.1 Checks built into the procedure

Every run performs input validation before the cascade and output validation before publishing. It returns five result sets per school year: the problems found; the distribution of students across rules with the change since the previous recorded run; and the three change summaries described in section 10. With the fail-on-validation flag set, an error blocks publishing and nothing after it runs.

Check	Catches
Workbook scenario has GIS block assignments under the same name	A rename on either side, which otherwise produces a scenario with almost no impact and reads as a finding.
One block-grade-scenario row in the GIS layer	A duplicate that would multiply a student and inflate a count.

Check	Catches
Every student assigned; count per scenario equals the roster	A rule gap, or a join that fanned out.
No student assigned to a school the scenario closes	A workbook that closes a building but leaves a program there.
The literal "neighborhood" never written as a school	A rule reading the workbook destination without the guard.
Next-year coverage, each side, by scenario and grade	Students the next-year rules did not place.
Feeder and crosswalk names match the roster	A spelling difference, or a destination that no longer has the program.
Immersion strand ends below its campus top grade with no feeder row	A K-5 strand at a K-8 building that the span rule would carry in place.
Scenario school names present in the roster	A GIS program name the normalization does not cover, which the change flag would read as a move.
Output table shape	A column added out of order, which select * would load silently into the wrong place.

Per-run counts by scenario and rule are written to an audit table keyed by run time, which is how the rule distribution is compared across runs and across procedure changes.

8.2 Comparison scripts

Four standalone queries sit alongside the procedure:

- **Forecast comparison.** Next-year counts by school, subgroup, and grade against the PRC-modified forecasts, as a full outer join. The two are not expected to agree, since the forecast applies cohort survival and in-migration. What the comparison catches is a school or strand present on one side only, which is almost always a scenario defined differently in the workbook and the forecast sheet, or a school spelled differently. This is the check that surfaced a partial "neighborhood" row emptying a program, and a feeder row pointing at a middle school that no longer takes that strand's 6th graders.
- **This year against next year.** Counts by school, group and grade, this year's assignment against the roll-forward, on both sides. In a steady-state school the two agree grade for grade; the entry grade of each level is where feeder and crosswalk problems show. A second result set lists every immersion student not simply carried forward in place, with the rule that placed them on each side.

- **Drill-down.** For a school and scenario: a name audit across all schools, the workbook and successor rows that mention the school, where its current students went and by which rule, and who lands there next year.
- **Feeder patterns by scenario.** For every immersion strand, source school and transition grade: where the students go next year under the status quo and under each scenario, with counts, and a column naming the scenarios whose pattern differs from the status quo. Read from the published table rather than from the feeder definition, so it shows what the rules did, including an unplaced strand. This is the view that surfaced the fifth feeder step.

8.3 Regression check

Before and after any change to the procedure, the published table is copied and compared with an EXCEPT in both directions on student, scenario, school, and group. A change that should only affect one rule, one scenario, or one feeder row is expected to show exactly that, and nothing else.

9. Outputs and how to read them

The published table holds one row per student per scenario, carrying the roster fields and the columns below.

Column	Meaning
assignment_school, Assignment_group, Assignment_note	This year's scenario assignment and the rule that made it.
living_2027	The school the student's address is zoned to under the 2027 set, at the student's grade.
scenario_es_living, scenario_ms_living, scenario_hs_living	The ES, MS, and HS boundary schools for the student's block under the scenario. These are what next-year results are summarized by ("scenario HS area").
NY_grade	Next year's grade; null for grade 12.
NY_Campus_Enrolled, NY_PRC_Group, NY_Campus_Note	Next year's status-quo school and group, and the rule that set them.
NY_Assignment_School, NY_Assignment_Group, NY_Assignment_Note	Next year's scenario school and group, and the rule that set them.
NY_change	1 if the scenario's next-year school differs from the status quo's; 0 if the same or both unplaced; null for grade 12.

This year's impact is the difference between assignment_school and campus_enrolled. Next year's impact is NY_change, and its sum over any demographic group is the number of students in that group the scenario moves. Both can be grouped by the student's demographics and by the scenario's attendance areas.

10. Summarizing the roll-forward

The roll-forward is incomplete. Next year's lottery results, special education placements, and hardship placements are not known when it runs, so the moves they will make are not in it (section 7). Within that limit it is faithful to the data PPS has: every placement follows a recorded rule, and the status quo and the scenarios are built from the same students in the same way.

It measures who is affected, not school composition. The demographic composition of schools under a scenario, as shown on the demographic pages, comes from this year's assignment (section 5), which places every current student as though the scenario took effect all at once. The roll-forward is for counting the students a scenario affects next year, overall, and by demographic group. It should not be read as any school's composition, or school by school. Transfers it cannot predict will move students in and out of individual schools.

The next-year results are reported as one measure, the share of continuing students the scenario moves: the sum of NY_change over the number of K-11 students, within whatever group the table is cut by. It is a share of students whose next-year school differs between the scenario and the status quo, built the same way on both model outputs (section 4). Students unplaced on both model outputs count as unmoved; students placed on one model output only count as moved. The procedure returns three summaries as its last three result sets, each a grid with the scenarios as columns, built from the same student rows it has just published, so the summaries always describe the run that produced them.

By high school region

Schools are grouped into four regions named for their high school pairs (Cleveland/Franklin, Ida B. Wells-Barnett/Lincoln, Grant/McDaniel, Jefferson/Roosevelt) plus Other. Each school is placed in the region where the plurality of the students it serves as a boundary school live, counted across the elementary, middle, and high school boundary columns of the roster under the 2027 set. Focus options and other non-boundary schools (da Vinci, Odyssey, Winterhaven, Richmond, MLC, ACCESS) are placed by hand. A student's region is the region of their status-quo next-year campus, so a 5th grader is counted in the region of the middle school they would attend next year under the status quo, not of the elementary school they are leaving.

By demographic group

Four metrics, each a partition of all students: direct certification for free meals (certified or not), special education (IEP or not), limited English proficiency (LEP or not), and race and ethnicity as recorded by the district, with a missing value reported as Unknown. An “All students” row carries the district rate, which is the baseline each group is read against. The question these tables answer is disproportionality: whether a scenario moves one group at a higher rate than it moves students overall.

By demographic group within region

The two summaries combined, with an All regions block and an All students row in each region, so any cell can be read against its region's overall rate and against the district's rate for that group. This is the table that distinguishes a scenario that moves one region heavily from one that moves one group heavily within a region.

Conventions shared by all three

- Scenario columns are built from the scenarios in the run; nothing in the procedure names one. The @SummaryScenarios parameter selects which appear, in what order, and under what presentation name, such as Scenario A.
- Rates for groups smaller than the @SummaryMinCell parameter (default ten) are suppressed, with the group size still shown so the reader can see the group exists and why the rate is blank.
- Each set describes one school year and leads with it; a two-year run returns a set per year rather than a blend. The workbook that carries them records the year in its title.
- The region definitions, the high school pairs and the hand-placed non-boundary schools, are tables inside the procedure, so a change to the regions is made in one place and applies to every summary.
- Alongside the three rate tables, the feeder-pattern grid (section 8.2) is the companion view for immersion: for each strand and transition, the school the students go to next year under the status quo and under each scenario.

11. Running the model

- Load the scenario workbook and the Closed School Successors tab (Python loaders), and draw or update the GIS boundary scenario with the Enrollment Balancing Toolbox. The scenario name must match exactly between the workbook and the GIS layer.
- Run the procedure. By default it models the latest roster year; it can be asked to model the preceding year in the same call. The first run after a change is made with validation set to report rather than block, so the result sets can be inspected.
- Read result set 1 (problems) and result set 2 (rule distribution and change since the last run) before using the output. A rule that suddenly places thousands more students, or drops to zero, is the signal that something moved.
- Result sets 3 to 5 are the change summaries. Pass @SummaryScenarios to choose and label the scenarios they show, and copy them into the summary workbook for that year.
- When the feeder table or crosswalk changes, rerun the this-year-against-next-year comparison for the affected strand.

Part 6. Floors, Capacity and Utilization

Each scenario is tested school by school against two limits, (1) an enrollment floor, tested against predicted 2031-2032 scenario enrollment, (2) the building's functional capacity, tested using each year's predicted scenario enrollment.

1. The thresholds

School type	Floor (minimum enrollment)	Ceiling
K–5	360	Functional capacity
K–8	500	Functional capacity
Middle school (6–8)	500	Functional capacity
High school (9–12)	1,100	Functional capacity

Floors are minimum student enrollments, not targets. A school may sit below its floor or above its capacity only temporarily, or where a scenario deliberately limits how many schools close. In scenarios with fewer consolidations, not every school reaches an enrollment above its floor. Functional capacity is the number of students a building can reasonably serve, based on the district's school utilization model.

2. How the tests are applied

- **School type comes from the modeled grades.** For each scenario, school, and year, the school type used is identified based on grade levels served in the scenario. A K–8 that a scenario converts to K–5 is tested as a K–5 from the conversion year on.
- **The floor is tested in 2031-2032.** Enrollment in other years is reported against the floor for information. The goal is to get to sustainable school sizes in five years.
- **Capacity is tested in each forecast year**, 2027-2028 through 2035-2036. A school with no functional capacity on file receives the floor test only. Testing each year helps ensure there is not overcrowding as PPS transitions to sustainable school sizes.
- **Anything that cannot be tested is reported as an error**, not passed, such as a grade span that the test does not recognize, or a group of students still enrolled at a school the scenario closes.
- **Utilization is modeled enrollment divided by functional capacity**. This is computed for PRC's baseline and for each scenario (Part 2, Section 4).

3. Outputs

The Floor Ceiling Detail table lists every school, scenario and year with its school type, enrollment, floor, functional capacity, utilization, seats available, and status. The Floor Ceiling Summary gives one row per school and scenario. Each row shows the school's type and enrollment in 2031-2032, whether scenario enrollment falls between its floor and its capacity, the result of the floor test, whether enrollment stays within capacity every year, the school's peak enrollment and the year of the peak, and the year the school closes if the scenario closes it.

Part 7. Assumptions, Limits and Sensitivity

The scenario numbers rest on measured data and on a small number of planning judgments. Part 7 describes the judgment calls, the limits of the data, and how a range of outcomes is produced.

1. Planning judgments

What the model assumes	Value in Scenarios A and B	Basis
When a school closes, most of its transfer students follow it to the school named to receive them.	At ten closing schools or grade spans, 66 to 90 percent of transfer students follow, and the rest return to the school for where they live. At the other nine, all of them follow.	Planning judgment
When an immersion program moves, most of its students move with it.	80 percent of immersion students follow at most programs. At a few programs the share is 70, 90 or 100 percent.	Planning judgment. The share is set lower where the new site is farther from the old one.
Resident families attend their new neighborhood school at the same rate as their current one (the capture rate).	PRC's forecast rate for the current school.	PRC's rate is carried over to the receiving school unchanged.
No more families leave PPS because of a closure or program move than PRC's forecast already expects.	No added loss of students. PRC's forecast already includes the share of students who leave PPS each year.	Assuming zero excess attrition provides an upper-bound stress test on a receiving school's functional capacity. If consolidation-induced attrition occurs, building utilization will be lower than modeled, so attrition alone will not push a school over capacity, though individual schools may experience additional downward pressure toward enrollment floors. The SEGC equity audit found that past immersion program moves did not appear to cause widespread departures from PPS (page 49).
Changes take effect in the year the scenario sets.	Closures and program moves take effect in full in that year. Boundary changes between two schools that both stay open apply to incoming students only.	District rule

The first two percentages in the table are the follow shares: the share of transfer and immersion students assumed to move with their school or program. They are not measured from past moves. For immersion programs, the district's SEGC equity audit (Green and Hanson, June 2026) found that when programs moved, students who stayed in PPS mostly moved with their program (pages 11 and 49). The audit does not give a percentage. PPS will review the follow shares, the 100 percent transfer rules and the capture-rate assumption with PRC before the Superintendent's final proposal to the Board.

2. Limits of the data

- **Census blocks are based on the 2010 geography**. The walking distances are an ArcGIS origin–destination list generated in 2016. Both are adequate for comparing scenarios school by school and year by year. Neither is a parcel-level description (Part 4, Sections 2 and 6).
- **The residency table describes where students live today**, over the three school years since the 2023-2024 K–8 boundary change. It does not include housing that is planned but not yet built. PRC's forecast does account for new housing in each school's total, but when a scenario divides a school's forecast among receiving schools, it uses the shares of where students live today. Students expected from new housing are split as if they lived where current students do.
- **Students without a usable address are carried as Undetermined.** These students have no block, so boundary counts and the student-level placement processes cannot assign them to a school. There are not many students without a usable address and their number is checked each year when we make October Enrollment reports.
- **Grade-level detail in a redistribution is an allocation, not a forecast.** Grade band totals are reliable, but the grade a returning student is shown in is not (Part 3).
- **The two models answer different questions.** The enrollment model applies percentage shares to PRC's forecast counts, so it can move part of a group. The demographic impact model places each of today's students at a single school and does not split groups. Their counts are not meant to match (Part 5, Section 1).

3. A range of outcomes

Each scenario is a single estimate. Because the follow shares are judgments, PPS produces a range for each receiving school by repeating the calculation with the shares set lower and higher, everything else held fixed, and reading each school's enrollment by year beside its floor and capacity. The example below is the Atkinson immersion program in Scenario A.

Atkinson immersion students following to Lent	30%	50%	80% (modeled)	100%
Lent, 2027-2028 (floor 360, capacity 798)	310	346	401	437
Atkinson, 2027-2028 (floor 360, capacity 519)	425	413	395	383

At a 30 percent follow share, Lent would hold 330 students in 2031-2032, below its floor. PPS modeled the scenario at 80 percent, as shown in the table. These figures replace the modeled share in the ledger directly. A full rerun differs by a few students because of rounding (Part 2, Section 3.7).

The exposure runs in both directions. Schools receiving a program are exposed to a low follow share, which can leave them below their floor. Neighborhood schools near capacity that receive a closed school's area are exposed to a high one. Schools that lose a program or transfer students carry the mirror image. PPS will publish the range for every receiving school and every year through 2031-2032 for the scenario the Superintendent recommends.

4. Review

PRC has reviewed the enrollment methodology in Part 2. Before adoption, the proposal will also be reviewed by district leadership and by the Board in public session. After adoption, PRC's own forecast, built on October 2026 enrollment and the adopted changes, becomes the forecast for staffing (Part 1, Section 2).

Appendix A. This-year cascade rules

Rules run in this order; each applies only to students not yet placed. The note written by each rule begins with its code.

Rule	Placement
R01	Immersion program moved by the workbook to a named school; whole strand follows.
R02	Program dissolved to neighborhood: fully reassigned, neighborhood the plurality destination; student goes to scenario boundary school as a resident.
R03	Resident whose block is zoned elsewhere under the scenario moves to that school.
R04	Resident or non-resident at a fully reassigned school goes to the scenario boundary school.
R05	A single workbook row fully reassigning the school and group to a named destination.
R06	Special or alternative school not in the scenario: no change.
R07	No boundary change and school not in the scenario: no change.
R08	Immersion program not in the scenario: no change.
R09	Special program not in the scenario: no change.
R10	Out-of-district student at a school the scenario does not affect: no change.
R11	Neighborhood student at a school neither closing nor changing boundaries: no change.

Rule	Placement
R12	Out-of-district student at a changing school follows the largest named destination.
R13	Transfer student whose school stays open: no change.
R14	Student with no in-district boundary at a school that stays open: no change.
R15	Non-resident at a school not on the neighborhood change list: no change.
R16	Resident whose workbook destination matches the GIS boundary school.
R17	Resident with no workbook row follows the GIS boundary school.
R18	Grades 9-12 continue at their current school (except MLC).
R19	Fully reassigned school and group go to the largest named destination.
R20	Resident already at their scenario boundary school: no change.
R21	Out-of-district student at a closing special school: largest named destination, else out of district.
R22	School and group not fully reassigned: no change.
R23	Immersion student with no workbook row: no change.
R24	Transfer student assigned to a school that itself closes is redirected to its primary successor.
R25	Fallback: fully reassigned, no destination; primary successor, else out of district.

Appendix B. Next-year rules

Each rule has a status-quo form (a) and a scenario form (b) unless noted. Notes begin with the rule code.

Rule	Placement
NY-01	Next year's grade = this year's grade + 1, grades K-11.
NY-02 / NY-03	Move up in place when next year's grade is within the school's grade span for the group (NY-02 status quo, NY-03 scenario). Non-residents, special programs and immersion inherit wider spans; a feeder row at the student's grade excludes them.
NY-04a / b	Immersion feeder pattern; scenario side requires the destination to still have the strand one grade up.

Rule	Placement
NY-04c	Scenario only: feeder destination's strand fully moved by the workbook; follow it.
NY-04d	Scenario only: the student's own school was moved by the scenario; follow the feeder pattern of the school attended today.
NY-04e	Scenario only: the strand at the student's own scenario school is fully moved by the workbook at next year's grade; follow it.
NY-05a / b / c	HS Spanish Immersion via the attendance-area crosswalk; c follows a program the workbook moved.
NY-06a / b	Grade 5, not immersion: middle school serving the address, as a resident.
NY-07a / b	Grade 8, not immersion: high school serving the address, as a resident.
NY-08	Change flag.

Appendix C. Reference tables

Immersion feeder patterns

These are today's feeder patterns, used for the status quo. A scenario changes them only through the modeling tool's workbook. In Scenarios A and B, for example, the workbook moves Ockley Green's Spanish Immersion program to George, so on the scenario side Beach's 5th graders, who feed Ockley Green today, go to George (Section 6.4, step 3).

From	Strand	Grade	To
Mt Tabor	Japanese Immersion	8 → 9	Grant
Hosford	Mandarin Immersion	8 → 9	Cleveland
Harrison Park	Mandarin Immersion	8 → 9	Cleveland
Harriet Tubman	Mandarin Immersion	8 → 9	Jefferson
Roseway Heights	Vietnamese Immersion	8 → 9	McDaniel
Brentwood	Russian Immersion	8 → 9	Franklin
Clark	Mandarin Immersion	5 → 6	Harrison Park
Woodstock	Mandarin Immersion	5 → 6	Harrison Park
MLK Jr	Mandarin Immersion	5 → 6	Harriet Tubman

From	Strand	Grade	To
Richmond	Japanese Immersion	5 → 6	Mt Tabor
Rose City Park	Vietnamese Immersion	5 → 6	Roseway Heights
Kelly	Russian Immersion	5 → 6	Brentwood
Lent	Spanish Immersion	5 → 6	Kellogg
Atkinson	Spanish Immersion	5 → 6	Kellogg
Bridger Creative Science	Spanish Immersion	5 → 6	Kellogg
Sitton	Spanish Immersion	5 → 6	George
James John	Spanish Immersion	5 → 6	George
Scott	Spanish Immersion	5 → 6	Roseway Heights
Rigler	Spanish Immersion	5 → 6	Beaumont
Beach	Spanish Immersion	5 → 6	Ockley Green
Ainsworth	Spanish Immersion	5 → 6	West Sylvan

Spanish Immersion middle school strands continue to their high school program through the attendance-area crosswalk below rather than a fixed feeder.

High school Spanish Immersion crosswalk

HS attendance area (hs_living_2027)	HS Spanish Immersion program
Cleveland	Franklin
Franklin	Franklin
Grant	McDaniel
Ida B. Wells-Barnett	Lincoln
Jefferson	Roosevelt
Lincoln	Lincoln
McDaniel	McDaniel
Roosevelt	Roosevelt